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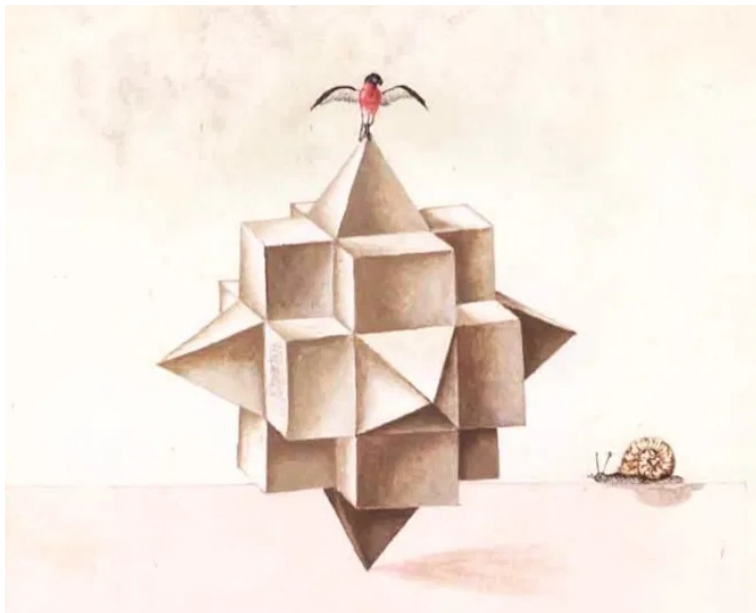
**SCHANUEL'S CLASSIFICATION
THEOREM FOR BOUNDED
POLYHEDRAL SETS**

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1 Introduction

1.1 Historical background

Over 2000 years ago the Greeks laid the basis for most of the mathematics that is still studied in school nowadays. Their love for beauty and conceptualization was most fertile ground for philosophy and mathematics: among the most celebrated concepts by Greek mathematicians and philosophers such as Euclid and Plato, lay the **polyhedra**. Their beauty and symmetry have for millennia captured the minds of countless mathematicians, architects and artists alike. To this day, polyhedra are objects of deep mathematical interest. Indeed, the set of solutions of a system of linear inequalities will generally form a polyhedron, and thus many problems in operational research amount to studying properties of a given polyhedron.

1.2 Motivation and Acknowledgements

Diophantus probably considered natural numbers not in the abstract way which we habitually now do, but as born from actual objects. While the method of formally adjoining negatives [...] leads to many results, we should not forget the objects themselves.

- William Lawvere, [10]

This work belongs to the field called *objective number theory* by W. Lawvere and S. Schanuel and the main references regarding this field are [14], [16], [15], [1], [5], [11], and [4]. The goal of objective number theory is to fix a category $\mathcal{C}$, consider the class of its objects $Ob(\mathcal{C})$ and try to describe - maybe even classify - the class $Ob(\mathcal{C})/\sim$ of objects up to isomorphisms of $\mathcal{C}$. Clearly this work - depending on $\mathcal{C}$'s complexity - can range from trivial to impossibly hard.

Example 1.2.1. Take as $\mathcal{C}$ the category of finite sets $\mathbf{Set}_f$. Since two finite sets are isomorphic if and only if they have the same cardinality, the class of its objects up to iso is just the set of natural numbers $\mathbb{N}$.

One of the main ideas to tackle this kind of problem is due to Grothendieck. If the category $\mathcal{C}$ comes equipped with a particularly well-behaved functor, this will force the objects of $\mathcal{C}$ to respect some sort of rigid algebraic structure, making it easier to study.

Example 1.2.2. Let X be a topological space and consider the category $\mathbf{Vect}(X)$ of vector bundles over X . This category admits two important functors: the direct sum $\oplus$ and the tensor product $\otimes$. These will result in two algebraic operations on the set $\mathbf{Vect}(X)/\sim$: a sum and a product. The resulting set of up-to-iso objects just about fails to be a ring, as it is only a monoid and not a group, but by tensoring with $\mathbb{Z}$ one can consider the natural *ring of vector bundles over a topological space X* , and use algebraic methods to study it. This ring is called the (topological) *K-theory* of X and much mathematical progress has been made in this direction.

The goal of objective number theory is to expand the program of Groethendieck's *Klassentheorie* to a non linear setting. In particular, in this thesis we will answer this problem for the category of bounded polyhedral sets.

1.3 Our problem and the tools needed

The main tool that will aid us in our problem is arguably the most renowned polyhedral invariant, the Euler characteristic χ . The key observation is that given two polyhedra P and Q it is the case that

$$\chi(P + Q) = \chi(P) + \chi(Q) \quad \text{and} \quad \chi(P \times Q) = \chi(P) \cdot \chi(Q)$$

meaning that the Euler characteristic preserves the algebraic structure of our category of polyhedra. Working with classical mathematical objects, such as polyhedra or - to a lesser extent - Euler characteristic, presents some foundational issues. As Dieudonné states in [3]:

... the mathematicians of the second half of the nineteenth century which were busy with these questions [of algebraic topology] speak freely of curves, of surfaces, of deformations, ..., without ever saying what they mean by these words.

This is especially true in our case, as we will see three different definitions of polyhedra and two of Euler characteristic. In particular, the Euler characteristic changed drastically in the last two centuries. It started as a discrete act of counting edges, vertices and faces and ended up setting the ground for the birth of topology. Today, many mathematicians and students only think of χ as the topological invariant, failing to see the connection with its original discrete formulation.

In our study topology, in particular connectedness, presents some difficulties and thus we will go back to the discrete definition of Euler characteristic. While doing geometry without topology presents some drawbacks, it also highlights some underlying information about polyhedra that is lost when considering continuous maps.

S. Schanuel studied the objective number theory problem for the category of bounded polyhedral sets, which are a generalization of the concept of polyhedra, in his paper [14]. In 2014, Schanuel passed away at the age of 81. His friend and collaborator W. Lawvere wrote [4]:

He gave his time and energy to the problems that presented themselves, he loved to discuss them and spin further solutions. He did not like to write, and seemed to be happy when scribbling and thinking.

Anyone who reads Schanuel's article [14] will realise that Lawvere was completely right. While the concepts presented are deep, the proofs are only sketched in extremely concise language and lack a sound approach. In this work we will provide the needed foundational work, leading to a rigorous proof of Schanuel's Theorem.

1.4 Outline

This thesis is structured in the following way.

Section 2 introduces the definition of polyhedra and Euler characteristic we will use throughout, paying particular attention to the lattice-theoretic nature of these objects. This section also includes key theorems that allow us to extend valuations such as χ from the simpler case of polytopes to the more complex setting of polyhedral sets.

Section 3 introduces extensive categories and the categorical tools needed to do objective number theory. It also provides a small generalization of subobject classifiers to be able to talk about partitions in a categorical sense.

Section 4 starts by defining polyhedral arrows, then goes on proving that the resulting category of polyhedral sets is extensive and thus satisfies the properties outlined in Section 3. We then define the Euler characteristic and dimension for every extensive category and we then use them to prove Schanuel's Theorem. As a corollary, we also get a complete classification of polyhedral sets.

2 The Euler characteristic and Bounded Polyhedra

2.1 Historical Background and Motivation

The concept of Euler Characteristic has been studied for a very long time. It was introduced¹ by Leonhard Euler in 1758 while describing (one of) his (many) famous equation(s), namely

Theorem 2.1.1 (Euler, 1758). *Any three-dimensional convex polyhedron P has Euler characteristic χ equal to*

$$\chi(P) = V - E + F = 2$$

where V , E , F are, respectively, the number of vertices, edges and faces of P .

From its discovery, this number kept on appearing in different forms in various area of geometry. To name a few:

Theorem 2.1.2 (Gauss-Bonnet, 1848). *Let M be a compact, two-dimensional Riemannian manifold. Let K be the curvature of M and k_g the geodesic curvature of ∂M . Then*

$$\int_M K dA + \int_{\partial M} k_g ds = 2\pi\chi(M)$$

Theorem 2.1.3 (Poincaré-Hopf, 1881). *Let M be a compact, oriented differential manifold, and let v be a vector field on M with isolated zeroes. Then*

$$i(v) = \int_M e(TM) = \chi(M)$$

where $i(v)$ is the total index of v , namely $i(v) = \sum_{p \in Z(v)} i(v, p)$, and $e(TM)$ is the Euler class of the tangent bundle.

Theorem 2.1.4 (Riemann-Hurwitz, 1857-1891). *Let X, Y be compact Riemann surfaces and let $f : X \rightarrow Y$ be a non constant holomorphic map with degree d . Then*

$$\chi(X) = d \cdot \chi(Y) - r_f$$

where r_f is the ramification index of f .

Of course, all these results use another - equivalent² - definition involving Betti numbers, namely:

¹Notably, Euler was not the first. Both René Descartes (1639) and Francesco Maurolico (1537) were aware of Theorem 2.1.1 in some equivalent form.

²Almost equivalent, as we will see in 2.4.2.

Definition 2.1.1 (Euler-Poincaré characteristic). *Let X be a differential manifold of dimension n . Then the Euler characteristic $\chi(X)$ is defined as*

$$\chi(X) = \sum_{i=0}^n (-1)^i b_i(X)$$

where $b_i(X)$, the i -th Betti number is the dimension of $H_{dR}^i(X)$ as an $\mathbb{R}$ -vector space.

In modern mathematics this last definition proved to be the most popular one, that naturally leads to the world of homological algebra and sheaves. The goal and reasoning, however, are pretty much the same: we wish to study some kind of mathematical *space* (a polyhedron, a differentiable manifold or a complex variety) by means of assigning them a *quantity* (Euler characteristic, a cohomology ring, or a Picard group). The underlying assumption being that quantities are easier to study because they only care about their operations. If the readers are interested in this philosophical point of view they can find out more in [9].

In this thesis we want to get back to the original setting in which this characteristic first arose. For us, the space will be a polyhedron, while the quantity will be the Euler characteristic. The aim of this thesis will be to see how far this “simple act of counting” can reach.

2.2 Definitions and first properties

Definition 2.2.1. *Let $S = \{x_1, \dots, x_k\}$ be a finite set of points in $\mathbb{R}^n$. The convex hull of S is called a **polytope**. Any set that can be written as a finite union of polytopes is called a **polyhedron**.*

Note that other sources may also refer to the former objects with the name polytopes and to the latter with the name *convex* polyhedra. Following the notation from [17], we will sometimes put a prefix $n-$ before a polytope or a polyhedron to specify its dimension. The dimension of a polytope is the dimension of the affine subspace it spans, while the dimension of a polyhedron is the maximum of the dimensions of the polytopes included in it.

Example 2.2.1. • A closed line segment is a 1-polytope.

- An n -dimensional simplex is an n -polytope.
- The empty set $\emptyset$ is conventionally regarded as a -1 -polytope.
- Any finite intersection of polytopes is itself a polytope.
- An open line segment is neither a polytope nor a polyhedron.
- A circle is neither a polytope nor a polyhedron.

- Every finite non-oriented graph is a polyhedron.
- Any polygon is a polyhedron.



- The text you're reading - being a finite union of square pixels - is a polyhedron.

Following the idea from before of using quantities to explore space, we need to define some sort of *quantitative measure* on our polytopes, to be then extended to polyhedra. The key concept here is that of *valuation*, that is an analogue to the concept of *measure* in a more general setting. How general this setting should be varies from one source to another, depending on the authors' use for it. In this thesis we will consider valuations on a *meet-semilattice*, that is a poset which admits all finite meets and a bottom element, called the zero. One can also dually define a join-semilattice, but throughout this thesis we will only consider their meet counterparts and thus when we will write "*semilattice*" what we actually mean is "*meet-semilattice*". In the same way, we will assume all our lattices to be distributive, unless specified otherwise.

Definition 2.2.2. *Let L be a lattice and S be a subsemilattice of L . A function $\nu : S \rightarrow \mathbb{R}$ is called a (real valued³) **valuation** if it satisfies*

$$P \leq Q \implies \nu(P) \leq \nu(Q) \quad (\text{Monotonicity})$$

$$\nu(0) = 0 \quad (\text{Normalisation})$$

$$\nu(P \wedge Q) + \nu(P \vee Q) = \nu(P) + \nu(Q) \quad (\text{Modularity Law})$$

whenever $P \vee Q$ is an element of S .

Example 2.2.2. • Let X be a finite set and let $(\mathcal{P}(X), \cap, \cup) = L = S$ be the powerset lattice. Then cardinality is a valuation.

- Let $(\Omega, \mathcal{A}, P)$ be a probability space. Then P is a valuation.
- Consider the semilattice of polytopes with dimension at most 2 (convex polygons, line segments and points) contained in the lattice of polyhedra with dimension at most 2 (polygons). Then perimeter is a valuation.

³In this thesis we will use $\mathbb{R}$ as the codomain of a valuation for historical reasons. What is really called for by the definition is a commutative monoid structure.

Example 2.2.3. The Euler-Poincaré characteristic satisfies the Modularity Law. Indeed, if P , Q and R are differential manifolds such that $R = P \cup Q$, we can write down the Mayer-Vietoris sequence

$$\begin{aligned} 0 \rightarrow H^0(R) &\rightarrow H^0(P) \oplus H^0(Q) \rightarrow H^0(P \cap Q) \rightarrow \\ &\hookrightarrow H^1(R) \rightarrow \dots \\ &\hookrightarrow H^k(R) \rightarrow H^k(P) \oplus H^k(Q) \rightarrow H^k(P \cap Q) \rightarrow 0 \end{aligned}$$

Now, the alternating sum of ranks for this sequence is clearly

$$\chi(R) - (\chi(P) + \chi(Q)) + \chi(P \cap Q)$$

But, by the Rank-Nullity Theorem, the alternating sum of ranks must be 0, and thus

$$\chi(P \cup Q) = \chi(P) + \chi(Q) - \chi(P \cap Q)$$

As observed in Example 2.2.1, the intersection of finitely many polytopes is itself a polytope, meaning that the set of polytopes contained in $\mathbb{R}^n$ has a semilattice structure.

Proposition 2.2.1. *Consider the semilattice of polytopes contained in $\mathbb{R}^n$. Then the function χ defined as*

$$\chi(P) = \begin{cases} 0 & \text{if } P = \emptyset \\ 1 & \text{otherwise} \end{cases}$$

*is a valuation, called **Euler characteristic**.*

Proof. The map χ is normalised and monotone by definition, so we only need to check the modularity law, which can be expressed as

Let P , Q be non-empty polytopes. Then, if $P \vee Q$ is a polytope, $P \wedge Q$ is non-empty.

By contradiction, assume that $P \wedge Q = \emptyset$. We can then consider the map $\sigma : P \vee Q \rightarrow [0, 1]$ given by

$$\sigma(x) = \begin{cases} 0 & \text{if } x \in P \\ 1 & \text{if } x \in Q \end{cases}$$

Since $P \wedge Q = \emptyset$, this map is both well defined and continuous (in the Euclidean topology). Now, take a pair of points $p \in P$ and $q \in Q$ and consider the line segment

$$\begin{aligned} \varphi : [0, 1] &\longrightarrow P \vee Q \\ t &\longmapsto tq + (1 - t)p \end{aligned}$$

Since $P \vee Q$ is convex, this map is both well defined and continuous, too. That means we can compose these two maps to get a map $\sigma \circ \varphi : [0, 1] \rightarrow [0, 1]$ which is continuous, has value 0 in 0 and 1 in 1. By the intermediate value theorem, there should be a $t \in [0, 1]$ such that $\sigma \circ \varphi(t) = 1/2$, and that's absurd. $\square$

Calling the map above Euler's characteristic might sound pretentious, but we will see in the next pages that this name is wholly justified.

Remark 2.2.1. The Euler characteristic defined above is a valuation with several special properties. In particular, it is

- continuous, with respect to the euclidean topology.
- homogeneous of degree⁴ 0, since

$$\chi(tP) = t^0 \chi(P)$$

- invariant with respect to isometry: if f is any isometry of $\mathbb{R}^n$, then $\chi(f(P)) = \chi(P)$.

It is worth noting that the valuations satisfying the properties above form a vector space. Indeed, any linear combination of (real-valued) valuations is still a valuation and it will satisfy the properties above if the generators satisfy them in the first place. Call this space V_n , where the n denotes the polytopes' dimension. One could ask: how big is V_n ?

Example 2.2.4. For example, for $n = 0$ the set Pt_0 has only two elements: $\{\emptyset, \mathbb{R}^0 = *\}$. Thus, there is only one possible non-null valuation, which is χ . So $\dim(V_0) = 1$.

For $n = 1$, we have two different valuations: the usual χ and the length of an interval (the 1-polytopes). Since the former is homogenous of degree 0 and the latter is of degree 1, these are in fact linearly independent. It's not difficult to see that these make up the whole space, so $\dim(V_1) = 2$.

We start to notice a pattern. The next question would be: is it true that $\dim(V_n) = n + 1$? The answer, while affirmative, is absolutely non-trivial, and it is part of a characterization theorem due to the Swiss mathematician Hugo Hadwiger.

Theorem 2.2.1 (Hadwiger, 1957). *Let $n \in \mathbb{N}$. Then*

$$\dim(V_n) = n + 1$$

Hadwiger then goes on to explicitly find a basis of V_n whose elements he calls *intrinsic volumes*, providing a full characterization theorem. The proof of this result, which can be found in [7], is long and difficult and it will not be discussed here. The interested reader can also find another, simpler, proof in [8], due to Daniel Klain.

⁴Abusing notation, we fix an origin O to treat our affine ambient space as a vector space so that the scalar multiplication is well-defined.

2.3 Extending valuations to a lattice

Hadwiger's deep result, while not particularly useful to us, is central in the field of convex geometry, and it has motivated mathematicians to extend valuations from a subsemilattice S to a whole lattice L containing S . The way to do that looks simple:

1. Take an element $Q \in L$ and write it as a finite join $Q = \bigvee P_i$ of elements $P_i \in S$.
2. Compute the valuation $\nu(Q)$ inductively using the Modularity Law. Namely: assume $\nu(P_0)$ and $\nu\left(\bigvee_{i \neq 0} P_i\right)$ are both well defined. Then we can extend ν to Q by defining

$$\nu(Q) = \nu\left(P_0 \vee \bigvee P_i\right) = \nu(P_0) + \nu\left(\bigvee P_i\right) - \nu\left(\bigvee (P_0 \wedge P_i)\right)$$

What's the catch? There are actually two problems with this construction. The first is that we must heavily rely on the group structure of $\mathbb{R}$ to be able to subtract $\nu\left(\bigvee (P_0 \wedge P_i)\right)$, making it almost useless in the case of valuations that take values in proper monoids. The second and most pressing problem is that to apply the reasoning above we must be able to write any element $Q \in L$ as a finite join of elements $P_i \in S$. That is only true if L is generated from S as its closure under finite join operations. Furthermore, even if our lattice was generated by S and the extension sketched above made sense, it is not immediately clear why this extension should be well defined. To prove it, we will need to introduce the well-known Inclusion Exclusion-Principle principle.

The Inclusion-Exclusion Principle consists of a historical technique⁵ used for counting the cardinality of finite sets. Namely, given a finite set $A = \bigcup A_i$, we can determine its cardinality by means of alternating sums:

$$\begin{aligned} \left| \bigcup_{i=1}^n A_i \right| &= \sum_i |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots \\ &= \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \left| \bigcap_{j \in J} A_j \right| \end{aligned}$$

Observe how the Inclusion-Exclusion Principle in the case $n = 2$ consists in just the modularity law for the cardinality valuation:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

This suggests us two things:

1. The Principle of Inclusion-Exclusion can be generalized to the lattice case by using valuations.

⁵First attributed to Abraham De Moivre (1718).

2. The Principle of Inclusion-Exclusion follows from the modularity law.

Definition 2.3.1. Let $\nu : S \rightarrow \mathbb{R}$ be a valuation. We say that ν satisfies the Inclusion-Exclusion Principle if

$$\nu\left(\bigvee_{i=1}^n A_i\right) = \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \nu\left(\bigwedge_{j \in J} A_j\right)$$

whenever $\bigvee A_i$ is an element of S . More generally, if ν 's codomain is only a commutative monoid, we can state this principle as

$$\nu\left(\bigvee_{i=1}^n A_i\right) + \sum_{\substack{J \subseteq \{1, \dots, n\}, \\ 0 \neq |J| \text{ even}}} \nu\left(\bigwedge_{j \in J} A_j\right) = \sum_{\substack{J \subseteq \{1, \dots, n\}, \\ |J| \text{ odd}}} \nu\left(\bigwedge_{j \in J} A_j\right)$$

Lemma 2.3.1. Any valuation $\nu : S \rightarrow \mathbb{R}$ satisfies the Inclusion-Exclusion Principle.

Proof. The proof is a simple induction on n . We've already seen how the base case is just the modularity law so we only need to check the inductive case. Assume that the thesis holds for $n-1$ and compute the join of n elements.

$$\begin{aligned} \nu\left(\bigvee_{1 \leq i \leq n} A_i\right) &= \nu\left(A_n \vee \bigvee_{1 \leq i \leq n-1} A_i\right) \\ &= \nu(A_n) + \nu\left(\bigvee_{1 \leq i \leq n-1} A_i\right) - \nu\left(A_n \wedge \bigvee_{1 \leq i \leq n-1} A_i\right) \\ &= \nu(A_n) + \sum_{J \subseteq \{1, \dots, n-1\}} (-1)^{|J|+1} \nu\left(\bigwedge_{j \in J} A_j\right) - \nu\left(\bigvee_{i \neq n} (A_i \wedge A_n)\right) \\ &= \nu(A_n) + \sum_{j \neq n} \nu(A_j) + \sum_{\substack{J \subseteq \{1, \dots, n-1\} \\ |J| > 1}} (-1)^{|J|+1} \nu\left(\bigwedge_{j \in J} A_j\right) + \\ &\quad + \sum_{J \subseteq \{1, \dots, n-1\}} (-1)^{|J|} \nu\left(\bigwedge_{j \in J} (A_j \wedge A_n)\right) \\ &= \sum_{1 \leq j \leq n} \nu(A_j) + \sum_{\substack{n \notin J \\ |J| > 1}} (-1)^{|J|+1} \nu\left(\bigwedge_{j \in J} A_j\right) + \sum_{\substack{n \in J \\ |J| > 1}} (-1)^{|J|+1} \nu\left(\bigwedge_{j \in J} A_j\right) \\ &= \sum_{\emptyset \neq J \subseteq \{1, \dots, n\}} (-1)^{|J|+1} \nu\left(\bigwedge_{j \in J} A_j\right) \end{aligned}$$

□

If one has a valuation that takes values in a proper commutative monoid instead of a group, Lemma 2.3.1 still holds and the proof is quite similar, so we

will omit it.

We will use this principle to prove the following theorem, due to Helmut Groemer.

Theorem 2.3.1 (H. Groemer, 1978). *Let L be a lattice, $S \subseteq L$ a subsemilattice, and assume L is generated by S through finite joins. If $\nu : S \rightarrow G$ is a valuation with values in an abelian group, then ν admits a unique extension to S^+ .*

Proof. To begin with, it's easier to think of S^+ as the lattice given by subsets of a certain set X . Stone's Prime Ideal Theorem allows us to make such an identification without any loss of generality, so for the rest of the proof we will treat S as a subclass of subset of X which is closed under intersections. Groemer's idea was to consider the set of characteristic functions for elements of S , namely, for every $P \in S$ define $P^* \in [P, \mathbb{Z}]$ by

$$P^*(x) = \begin{cases} 1 & \text{if } x \in P \\ 0 & \text{otherwise} \end{cases}$$

The key observation is that the operation $(-)^*$ satisfies the Modularity Law (and thus the Inclusion-Exclusion Principle), too. Indeed, if $P_1 \vee P_2$ is an element of S , we have

$$(P_1 \vee P_2)^* = P_1^* + P_2^* - (P_1 \wedge P_2)^*$$

and thus, if $\bigvee P_i \in S$,

$$\left(\bigvee_1^n P_i \right)^* = \sum_1^n P_i^* - \sum_{1 \leq i < j \leq n} (P_i \wedge P_j)^* + \dots + (-1)^{n+1} \left(\bigwedge_1^n P_i \right)^*$$

We will prove our result by taking a small detour and firstly prove the following claim.

Let P_i be elements of S and b_i be integers. If

$$b_1 P_1^* + \dots + b_n P_n^* = 0 \quad \text{then} \quad b_1 \nu(P_1) + \dots + b_n \nu(P_n) = 0$$

By contradiction, let's assume there exist $P_1 \dots P_k$ and integers $b_1, \dots, b_k$ for which the claim is false, i.e.

$$b_1 P_1^* + \dots + b_n P_n^* = 0 \quad \text{but} \quad b_1 \nu(P_1) + \dots + b_n \nu(P_n) \neq 0$$

Now consider the meet semilattice generated by $P_1, \dots, P_k$:

$$\begin{aligned} D_1 &= P_1, \dots, D_k = P_k \\ D_{k+1} &= P_1 \wedge P_2, \dots, D_{k'} = P_{k-1} \wedge P_k \\ &\vdots \\ D_{2^k-1} &= \bigwedge_{1 \leq i \leq k} P_i \end{aligned}$$

Note how, for every n, m the meet $D_m \wedge D_n$ equals some D_t with $t \geq \max(n, m)$. Now, there clearly exist $r \in \mathbb{N}$ and $(c_r, \dots, c_{2^k-1}) \in \mathbb{Z}^{2^k-r}$ such that

$$\sum_{r \leq i \leq 2^k-1} c_i D_i^* = 0 \quad \text{but} \quad \sum_{r \leq i \leq 2^k-1} c_i \nu(D_i) \neq 0 \quad \text{and} \quad c_r \neq 0 \quad (1)$$

because we can just put $c_i = b_i$ for $i \leq k$ and $c_i = 0$ for $i > k$. In this example r would be the first i such that $b_i \neq 0$. Since we know that there is at least one such r and that it can be at most $2^k - 1$, then we can consider s as the maximum number such that the request above is satisfied. Now, what can we say about the set D_s ? Take an element $x \in D_s$. Since $D_s^*(x) = 1$, we cannot have that $D_i^*(x) = 0$ for every $i > s$ because otherwise $\sum_{i \geq s} D_i^* \neq 0$. So every point of D_s is in some other D_i for some $i > s$. Thus

$$\begin{aligned} D_s &= D_s \wedge (D_{s+1} \vee \dots \vee D_{2^k-1}) \\ &= (D_s \wedge D_{s+1}) \vee \dots \vee (D_s \wedge D_{2^k-1}) \\ \implies \nu(D_s) &= \sum \nu(D_s \wedge D_i) - \sum \nu(D_s \wedge D_i \wedge D_j) + \dots \\ \implies (D_s)^* &= \sum (D_s \wedge D_i)^* - \sum (D_s \wedge D_i \wedge D_j)^* + \dots \end{aligned}$$

But now, since each of this finite meet is contained in some other D_t , we get that there exists some integer coefficients d_t such that

$$\nu(D_s) = \sum_{t > s} d_t \nu(D_t) \quad \text{and} \quad D_s^* = \sum_{t > s} d_t D_t^*$$

Substituting these identities in Equation 1 gives us a similar equation starting from the $s + 1$ -th term, contradicting the maximality of s .

Now we can finally prove the extension result. Take an element $P \in S^+$ than can be written in two different ways as a join of elements P_i and P'_i in S . By the Inclusion-Exclusion Principle, we can write

$$\begin{aligned} P &= \bigvee P_i \implies \\ P^* &= \sum P_i^* - \sum P_i \wedge P_j + \dots = \sum_i b_i Q_i^* \end{aligned}$$

We extend ν to S^+ by defining

$$\nu(P) = \sum_i b_i \nu(Q_i)$$

Now, clearly the other join decomposition $P = \bigvee P'_i$ will yield another expression $P^* = \sum c_i R_i^*$, which will define another way to express $\nu(P)$. We will denote this with $\bar{\nu}$:

$$\bar{\nu}(P) = \sum_i c_i \nu(R_i)$$

But now, since $P^* = P^*$, we have

$$\begin{aligned} 0 &= P^* - P^* = \sum_i b_i Q_i^* - \sum_i c_i R_i^* \\ \implies 0 &= \sum_i b_i \nu(Q_i) - \sum_i c_i \nu(R_i) = \nu(P) - \bar{\nu}(P) \end{aligned}$$

by the claim. Thus ν can be extended to S^+ . The uniqueness of the extension is given by the fact that such an extension has to respect the Inclusion-Exclusion Principle. $\square$

Remark 2.3.1. Theorem 2.3.1 is formulated slightly differently from its original appearance [6] and from the later book [13]: instead of valuations in an abelian group, they directly consider a real vector space and define an “integral” with respect to the chosen valuation. This was made in an effort to pursue Gian-Carlo Rota’s program to “linearise” the theory of distributive lattices, in a way similar to how cohomology groups linearise the theory of smooth manifolds. For our purposes only integer combinations are needed.

Now, to recover the known Euler Characteristic, one has to talk about *systems of faces* for a polyhedron.

Definition 2.3.2. We say that a polytope P' is a face of another polytope P if there exists an hyperplane H such that $P' = P \cap H$ and P is wholly contained in one of the half-spaces determined by H .

Definition 2.3.3. Given a polyhedron P , we say that a collection of polytopes F is a **system of faces** for P if

1. $\bigvee_{P_i \in F} P_i = P$.
2. $\emptyset \in F$.
3. If $P_i, P_j \in F$, then $P_i \wedge P_j$ is a face of both P_i and P_j . Moreover $P_i \wedge P_j \in F$.

By applying Theorem 2.3.1 we can extend the Euler valuation that equals 1 on every non-empty polytope to the lattice of polyhedron. The result is the classical Euler characteristic, hence the name.

Theorem 2.3.2 (Euler’s formula for polyhedra). *Let F be a system of faces for a polyhedron P and denote with f_i the number of elements in F of dimension i . Then*

$$\chi(P) = f_0 - f_1 + f_2 - \dots$$

The proof of this theorem involves tools not discussed in this thesis, and so we will omit it. The interested reader can find the proof in [13], Theorem 5.2.4.

2.4 Extending valuations to a Boolean algebra

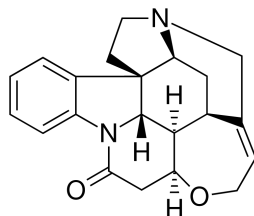
We successfully extended valuations from a semi-lattice to a whole lattice, so why not go even further and extend them to a Boolean algebra?

Definition 2.4.1. Any subset $P \subseteq \mathbb{R}^n$ that can be written as a (finite) Boolean combination of polytopes is called a **polyhedral set**. Meaning:

1. Every polytope is a polyhedral set.
2. Finite intersections and unions of polyhedral sets are polyhedral sets.
3. The complement of a polyhedral set is a polyhedral set.

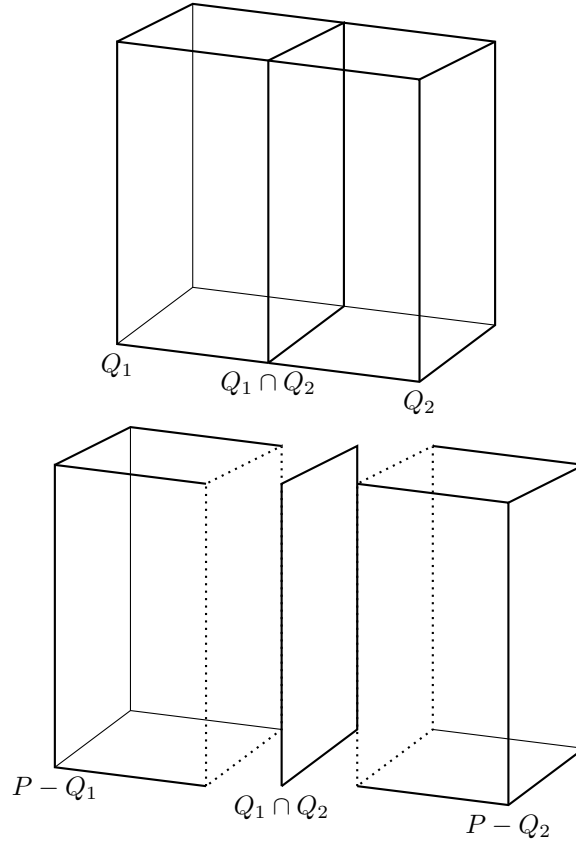
Example 2.4.1. • Every polyhedron is a polyhedral set.

- Every affine space $\mathbb{R}^n$ is a polyhedral set, being the complement of the polytope $\emptyset$.
- An open line segment is a polyhedral set.
- A circle is not a polyhedral set.
- The open ray $r = \{x \in \mathbb{R} \mid x > 0\}$ is not a polyhedral set. Indeed, every unbounded polyhedral set must be either the complementary set of a bounded polyhedron or of an unbounded polyhedron with strictly lesser dimension. The ray r is neither so it cannot be a polyhedral set.
- Every structural formula for a chemical compound is a polyhedral set.



The structural formula for Strychnine. By [Vaccinationist - PubChem](#),
[Public Domain](#)

The main appeal behind extending our lattice of polyhedra to a Boolean algebra is the fact that we don't need to engage with overlaps anymore. Indeed, if we need to evaluate the Euler characteristic of a simple polyhedron (and we successfully extended χ to all polyhedral sets) $P = Q_1 \cup Q_2$, we can just write P as the *disjoint* union of three different polyhedral sets: $P - Q_1$, $P - Q_2$ and $Q_1 \cap Q_2$.



Remarkably, the Euler characteristic is still preserved, ignoring the fact that we just disconnected a previously connected object.

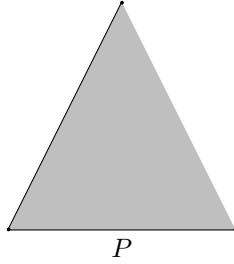
Remark 2.4.1. We can view our Boolean algebra as a small category, with arrows $P \rightarrow Q$ given by the order $P \leq Q$. In this category, sums (coproducts) are given by joins. If we separate each polyhedral set in disjoint unions as above, the modularity law simply becomes

$$\chi((P - Q_1) + (P - Q_2) + (Q_1 \cap Q_2)) = \chi(P - Q_1) + \chi(P - Q_2) + \chi(Q_1 \cap Q_2)$$

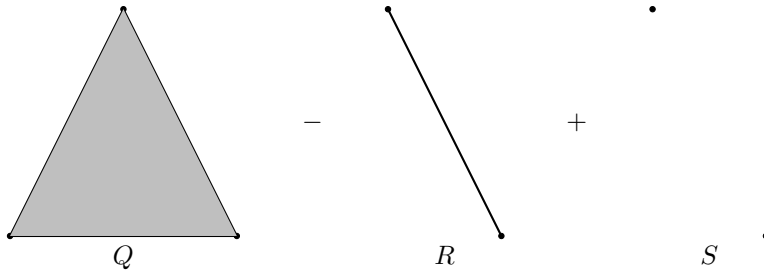
So the Euler characteristic behaves in a way similar to a group homomorphism. This, of course, makes no sense since the Boolean algebra of polyhedral sets is not a group, but it's an intuition that we will develop later.

Intuitively speaking, what is the naive method of extending χ ?

Example 2.4.2. Let's say we need to define the Euler characteristic for the polyhedral set sketched below, a triangle missing an open edge, in a way that makes sense, i.e. that agrees with Theorem 2.3.2.



One way to do this would be to write our polyhedral set as the join of differences between objects for which the Euler characteristic is known:



And then adding and subtracting the known characteristics.

$$\chi(P) := \chi(Q) - \chi(R) + \chi(S)$$

The reader is invited to check that the extension above agrees with Theorem 2.3.2.

There is one small problem with this sketch of construction, that is, differences between (polyhedral) sets are not the same as differences between integers. For example, in the calculation above we could have considered $R' = R + \{*\}$, where $\{*\}$ is a point not contained in Q . Then

$$P = Q - R' + S \quad \text{but} \quad \chi(P) = \chi(Q) - \chi(R') + \chi(S) + 1$$

Thus, not only do we need joins to be disjoint, but we also need differences (i.e. meets with negations) to be proper: we cannot subtract from a polyhedron parts that it doesn't have. That being said, this is only an impediment to χ bearing a similarity to a group homomorphism and not to the possibility of extending χ .

Theorem 2.4.1. *Let $\nu : L \rightarrow G$ be a valuation with values in an abelian group, and denote by L^- the Boolean algebra freely generated by L . Then ν admits a unique extension to L^- .*

Proof. The proof is basically the same as that of Theorem 2.3.1. First observe that, being L^- a Boolean algebra, we can express a polyhedral set P in disjunctive normal form

$$P = \bigvee_i (Q_i - R_i)$$

where it is not restrictive to limit ourselves to the case where $R_i \leq Q_i$. Consider the sub-poset $L^\times$ given by the elements of L^- that can be expressed as binary differences $(X - Y)$ with $Y \leq X$, $X, Y \in L$. It is not difficult to see that $L^\times$ is a sub-semilattice which generates L^- . Indeed,

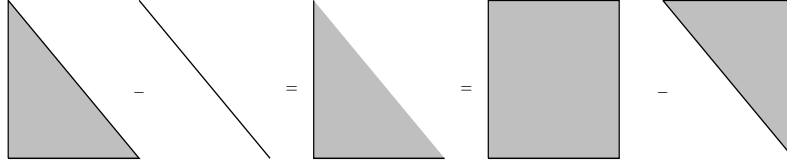
$$(X - Y) \wedge (A - B) = \underbrace{(X \wedge A)}_{\in L} - \underbrace{(Y \wedge B)}_{\in L}$$

and moreover $\emptyset \in L^\times$, as $\emptyset = X - X$. Thus, if we were to extend ν to elements of the form $X - Y$, we can use Theorem 2.3.1 to extend it to the whole L^- . The way to do that is just as in Example 2.4.2 by putting

$$\nu(Q - R) = \nu(Q) - \nu(R)$$

We only need to prove that this definition is well posed, that is, every time that $X - X' = Y - Y'$ we also must have

$$\nu(X) - \nu(X') = \nu(Y) - \nu(Y')$$



Call $Z = X - X' = Y - Y'$, with $X' \leq X$ and $Y' \leq Y$. Now, since

$$X = Z \vee X' \quad \text{and} \quad Y = Z \vee Y'$$

We get

$$X \vee Y' = Z \vee X' \vee Y' = Z \vee Y' \vee X' = Y \vee X'$$

Moreover, since

$$Z \wedge X' = 0 = Z \wedge Y'$$

we also have that

$$X \wedge Y' = (Z \vee X') \wedge Y' = (Z \wedge Y') \vee (X' \wedge Y') = X' \wedge Y'$$

Analogously $Y \wedge X' = X' \wedge Y'$. What's left is just a Modularity Law application:

$$\begin{aligned} \nu(X \vee Y') &= \nu(X) + \nu(Y') - \nu(X \wedge Y') = \nu(X) + \nu(Y') - \nu(X' \wedge Y') \\ \nu(Y \vee X') &= \nu(Y) + \nu(X') - \nu(Y \wedge X') = \nu(Y) + \nu(X') - \nu(X' \wedge Y') \end{aligned}$$

But, since $X \vee Y' = Y \vee X'$, the terms on the right must also be equal. That is,

$$\nu(X) + \nu(Y') = \nu(Y) + \nu(X') \implies \nu(X) - \nu(X') = \nu(Y) - \nu(Y')$$

Thus ending our proof. $\square$

Corollary 2.4.1. *If P is an open⁶ bounded convex polyhedral set of dimension k , then*

$$\chi(P) = (-1)^k$$

Proof. The proof relies on Theorem 2.4.1 in conjunction with Theorem 2.3.2. Let P be an open bounded convex polyhedral set of dimension k and let $\bar{P}$ be its closure. Since $\bar{P}$ is closed, bounded and convex, it's actually a polytope and its Euler characteristic is by definition $\chi(\bar{P}) = 1$. By Theorem 2.3.2 we can write

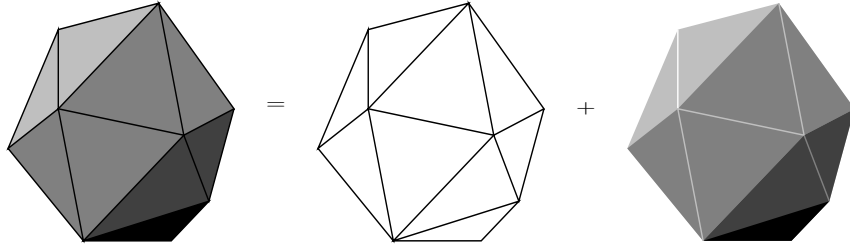
$$1 = \chi(\bar{P}) = f_0 - f_1 + f_2 + \dots + (-1)^k f_k$$

Now, f_k must be equal to 1 since there is only one k -dimensional face, being $\bar{P}$ itself. Thus the Euler characteristic of $\bar{P}$'s *boundary* must be given by the faces with dimension strictly less than k . This means that

$$\chi(\partial\bar{P}) = \chi(\partial P) = \chi(\bar{P}) - (-1)^k$$

Now, by Theorem 2.4.1 we can compute the Euler characteristic of P by

$$1 = \chi(\bar{P}) = \chi(P) + \chi(\partial P) = \chi(P) + 1 - (-1)^k$$



And thus

$$\chi(P) = (-1)^k$$

□

Remark 2.4.2. We would like to call the reader's attention to how this version of the Euler characteristic is fundamentally different from the one involving Betti numbers in algebraic topology. In fact, the cohomology groups are homotopy invariants and so is the Euler-Poincaré characteristic. By the Poincaré Lemma, each n -th dimensional open cube $(0, 1)^n$ has Euler-Poincaré characteristic 1, as H^0 is its only non-trivial cohomology group. However, in our definition each n -open-cube - being an open convex polyhedral set - has Euler characteristic

$$\chi((0, 1)^n) = (-1)^n$$

⁶In the affine space it spans.

2.5 Products of Polyhedral sets

The intuition behind Remark 2.4.1 can be expanded, leading to another important property of the Euler characteristic, often ignored. Indeed, disjoint unions are sums not only in a Boolean algebra, but also in **Set** and **Top** which come equipped with a Cartesian product too. Is it possible that χ could behave similarly to a *ring* morphism too?

Proposition 2.5.1. *Let P and Q be polyhedral sets. Then the Cartesian product $P \times Q$ is a polyhedral set and the following holds*

$$\chi(P \times Q) = \chi(P) \cdot \chi(Q)$$

Proof. We will prove the first statement by induction on the construction of P and Q : let us firstly consider the case where P and Q are convex hulls of finite sets. To fix notation:

$$\begin{aligned} P' &= \{p_1, \dots, p_n\} & Q' &= \{q_1, \dots, q_m\} \\ P &= \text{conv}(P') & Q &= \text{conv}(Q') \end{aligned}$$

We claim that $P \times Q = \text{conv}(P' \times Q')$. Indeed, given an element z in $\text{conv}(P' \times Q')$ we can write it as⁷ $z = \sum_{i,j} \lambda_{ij}(p_i + q_j)$ where $\lambda_{ij} \in [0, 1]$ and $\sum_{i,j} \lambda_{ij} = 1$. We then have

$$z = \sum_{i,j} \lambda_{ij} p_i + \sum_{i,j} \lambda_{ij} q_j = \sum_i p_i \sum_j \lambda_{ij} + \sum_j q_j \sum_i \lambda_{ij}$$

Now, by putting $\lambda_{i-} = \sum_j \lambda_{ij}$ and $\lambda_{-j} = \sum_i \lambda_{ij}$, we obtain

$$z = \sum_i \lambda_{i-} p_i + \sum_j \lambda_{-j} q_j$$

Clearly, $\sum_i \lambda_{i-} = \sum_j \lambda_{-j} = \sum_{i,j} \lambda_{ij} = 1$ and thus z is an element of $\text{conv}(P') \times \text{conv}(Q') = P \times Q$. Conversely, if z is an element of $P \times Q$, say $z = \sum_i \lambda_i p_i + \sum_j \gamma_j q_j$, we clearly see that

$$1 = 1 \cdot 1 = \left(\sum_i \lambda_i \right) \cdot \left(\sum_j \gamma_j \right) = \sum_{i,j} \lambda_i \gamma_j$$

⁷We are introducing here a minor notation abuse by identifying the points p_i and q_j as their images through an embedding in a Euclidean space of sufficiently high dimension. This is done by adding a streak of zeroes before the q_j s and after the p_i s.

Which means

$$\begin{aligned}
\sum_{i,j} \lambda_i \gamma_j (p_i + q_j) &= \sum_{i,j} \lambda_i \gamma_j p_i + \sum_{i,j} \lambda_i \gamma_j q_j \\
&= \sum_i \lambda_i p_i \left(\sum_j \gamma_j \right) + \sum_j \gamma_j q_j \left(\sum_i \lambda_i \right) \\
&= \sum_i \lambda_i p_i + \sum_j \gamma_j q_j \\
&= z
\end{aligned}$$

So the thesis holds for polytopes. In order to extend our result to polyhedral sets we can observe that

$$\begin{aligned}
(P \times R) \cap (Q \times R) &= (P \cap Q) \times R \\
(P \times R) \cup (Q \times R) &= (P \cup Q) \times R
\end{aligned}$$

Negation is a bit less obvious: we will use that

$$(P \times R)^c \cap (\mathbb{R}^d \times R) = P^c \times R$$

where d is the dimension of P . The formula above can hopefully be more clear when restated in logical terms: let $t(x, y)$ be a term in the variables x and y . Being an element of the set on the left means

$$\begin{aligned}
&\forall x \, t(x, y) \text{ cannot satisfy the property } P \text{ OR } \forall y \, t(x, y) \text{ cannot satisfy } R. \\
&\text{Moreover } \exists x \text{ such that } t(x, y) \text{ satisfies } R.
\end{aligned}$$

which is clearly equivalent to

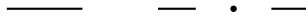
$$\exists x \text{ such that } t(x, y) \text{ satisfies } R \text{ but } \forall y \, t(x, y) \text{ does not satisfy } P.$$

which means being an element of the set on the right.

To prove the second - and perhaps more interesting - statement, we assign to each bounded polyhedral set P a polynomial $\varphi_P(X)$ in the following way.

1. Decompose P in a disjoint union of bounded open convex polytopes.
2. Define $a_i \in \mathbb{N}$ to be the number of bounded open convex polytopes with dimension i occurring in our decomposition.
3. Define $\varphi_P(X) = \sum a_i X^i$

Now, this assignment is *not* well defined. For example, take the open interval $P = (0, 1)$. We can decompose it as just itself or as two open intervals with a point.



Clearly, these decompositions yield two different polynomials,

$$\varphi_{P_1}(X) = X \quad \text{and} \quad \varphi_{P_2}(X) = 2X + 1$$

The important thing is that evaluating any of these polynomials in -1 will result in $\chi(P)$, independently from the chosen decomposition. This follows immediately from Remark 2.4.1 and Corollary 2.4.1. The thesis now follow from the observation that

$$\chi(P \times Q) = \varphi_{P \times Q}(-1) = \varphi_P(-1) \cdot \varphi_Q(-1) = \chi(P) \cdot \chi(Q)$$

where the equality

$$\varphi_{P \times Q} = \varphi_P \cdot \varphi_Q$$

is given by the fact that **Set** is a *distributive* category. That is, for every set A , B , C we have

$$A \times (B + C) = A \times B + A \times C$$

Thus, if - say - we have a_n (resp. b_m) different bounded open convex polyhedral sets of dimension n (resp. m) in the decomposition of P (resp. Q), then

$$\left(\sum_i^{a_n} P_i \right) \times \left(\sum_j^{b_m} Q_j \right) = \sum_i^{a_n} \sum_j^{b_m} P_i \times Q_j$$

Since the dimension of $P_i \times Q_j$ is $n + m$, we end up with $a_n + b_m$ terms of dimension $n + m$ in the decomposition of $P \times Q$. By considering each possible term, we end up with the polynomial equality

$$\varphi_{P \times Q} = \varphi_P \cdot \varphi_Q$$

concluding our proof. □

3 Extensive Categories

3.1 Goals and Acknowledgements

Before beginning this chapter we would like to bring the reader's attention to the last step in Proposition 2.5.1. The key concept that ended the proof was the distributive property of **Set**, that translated in a distributive property of polynomials. Truth be told, the real property that acted behind the scenes is a distributive property for the category of polyhedral sets. Of course, this sentence makes no sense as we didn't say what a morphism of polyhedral sets should be, but we shall define it in due time. The goal of this next section is to talk about the categorical machinery that makes a polynomial argument such as the proof of Proposition 2.5.1 possible, in the most general setting possible. Lastly, this whole section is taken more or less entirely from [2], which shall be the reference for all the following theorems unless stated otherwise.

3.2 Definitions and Examples

Definition 3.2.1. *A category $\mathcal{D}$ with finite products and sums is called distributive if the canonical map*

$$\delta : A \times B + A \times C \longrightarrow A \times (B + C)$$

is an isomorphism, for each object A , B , and C .

It should be noted that here the terms “sums” and “coproducts” are synonymous and the terms will be used interchangeably throughout this thesis.

Example 3.2.1. Every cartesian closed category possessing finite sums is distributive, and thus so is every topos. This is because the functor $A \times -$ is a left adjoint to $(-)^A$, and thus preserves colimits such as sums.

Example 3.2.2. The category of vector spaces is not distributive. For the distributivity condition would imply that

$$A + (B + C) = A \times (B + C) = A \times B + A \times C = A + B + A + C = 2A + B + C$$

and thus $\dim(A) = 0$ for every vector space A .

Definition 3.2.2. *A category $\mathcal{E}$ with finite sums is called **extensive** if the canonical functor*

$$+ : (\mathcal{E}/A) \times (\mathcal{E}/B) \longrightarrow \mathcal{E}/(A + B)$$

is an equivalence, for any two objects A and B . If $\mathcal{E}$ also has finite products, it is called preextensive; and if it has all finite limits, it is called lexextensive.

The main property we're interested in is *extensivity* and we shall eventually prove that a preextensive category is also distributive.

Let us spend a few words on Definition 3.2.2. The essential surjectivity of $+$ spelled out basically says that every time an object has an arrow into a sum

$$A \xrightarrow{f} B + C$$

the first object is actually (isomorphic to) a sum and the arrow is (isomorphic to) a sum of parallel arrows.

$$\begin{array}{ccc} A_1 & \xrightarrow{f_1} & B \\ \iota_1 \downarrow & & \downarrow \iota_B \\ A_1 + A_2 & \xrightarrow{f_1 + f_2} & B + C \\ \iota_2 \uparrow & & \uparrow \iota_C \\ A_2 & \xrightarrow{f_2} & C \end{array}$$

The fullness of $+$ consists in a similar “splitting” property: whenever one has a commutative diagram of the following form

$$\begin{array}{ccc} A + B & \xrightarrow{\varphi} & A' + B' \\ & \searrow a+b \quad \swarrow a'+b' & \\ & X + Y & \end{array}$$

Then φ splits in $\varphi_A + \varphi_B$ so that

$$a' \circ \varphi_A = a \quad \text{and} \quad b' \circ \varphi_B = b'$$

This is fundamentally a property of good behaviour for sums and it's not immediately clear why it should imply any properties of limits such as products. We shall see that this happens because extensivity implies the existence of a certain type of pullbacks.

First, it should be useful to see some examples of extensive categories.

Example 3.2.3. Set is extensive.

To see this, take a function $f : A \rightarrow B + C$ and consider the subsets $f^{-1}(B)$ and $f^{-1}(C)$ of A . Since B and C are disjoint, $f^{-1}(B)$ and $f^{-1}(C)$ are disjoint too, so that we have

$$A = f^{-1}(B) + f^{-1}(C)$$

Moreover, it's clear that

$$f = f \upharpoonright_{f^{-1}(B)} + f \upharpoonright_{f^{-1}(C)}$$

To check fullness, start by taking a commutative diagram

$$\begin{array}{ccc} A \sqcup B & \xrightarrow{\varphi} & A' \sqcup B' \\ & \searrow \alpha+\beta \quad \swarrow \alpha'+\beta' & \\ & X \sqcup Y & \end{array}$$

As before, φ splits in $\varphi = \varphi \upharpoonright_A + \varphi \upharpoonright_B$. The fact that the codomain of $\varphi \upharpoonright_A$ is exactly A' and that the same is true for $\varphi \upharpoonright_B$ is just a consequence of the diagram's commutativity.
So **Set** is indeed extensive.

Example 3.2.4. Let X be a finite, non-empty set. Then the power set $\mathcal{P}(X)$ viewed as a poset category is not extensive.

First, let us observe that to have an arrow $A \rightarrow B + C$ means to have $A \subseteq B \cup C$. Is it true that A can be viewed as a sum of two sets A_1 and A_2 such that the square above commutes? Certainly. Take $A_1 = A \cap B$ and $A_2 = A \cap C$: then we have

$$\begin{array}{ccc} A_1 = A \cap B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A = A_1 \cup A_2 & \longrightarrow & B \cup C \\ \uparrow & & \uparrow \\ A_2 = A \cap C & \longrightarrow & C \end{array}$$

This time the problem lies not in the essential surjectivity, but in the fullness. Indeed, if one has

$$\begin{array}{ccc} A \cup B & \longrightarrow & A' \cup B' \\ & \searrow \quad \swarrow & \\ & X \cup Y & \end{array}$$

with $A, A' \subseteq X$, $B, B' \subseteq Y$ and $A \cup B \subseteq A' \cup B'$, it needs not be that

$$A \subseteq A' \quad \text{and} \quad B \subseteq B'$$

For example, set $X = Y = B' = \{0, 1\}$, $A = \{0\}$, $B = \{1\}$ and $A' = \emptyset$.

The fundamental property that makes **Set** extensive while $\mathcal{P}(X)$ fails to be so, is that sums in **Set** are *disjoint*, while in $\mathcal{P}(X)$ they are not. Truly, we only started speaking about sums of polyhedra when we made it possible to disjoin them. We will refine this intuition more precisely after introducing some basic properties of extensive categories.

3.3 Fundamental properties of Extensive Categories

The attentive reader would have noticed how we didn't mention faithfulness neither to develop an intuition of it nor to check it to verify that **Set** is extensive in Example 3.2.3. This is because of the following lemma.

Lemma 3.3.1. *The functor $+$ is always faithful.*

Proof. Indeed, let $f, f' : A \rightarrow C$ and $g, g' : B \rightarrow D$ such that

$$f + g = f' + g'$$

Then we have

$$\begin{array}{ccccc}
A & \xrightarrow{f} & C & & \\
\downarrow \iota_A & \searrow & \swarrow \iota_C & & \\
A+B & \xrightarrow{f+g} & C+D & & \\
\parallel & & \parallel & & \\
A+B & \xrightarrow{f'+g'} & C+D & & \\
\uparrow \iota_A & \swarrow & \searrow \iota_C & & \\
A & \xrightarrow{f'} & C & &
\end{array}$$

The internal square commutes by hypothesis, while the other squares commute by definition. Thus the external square commutes and $f = f'$. Similarly, $g = g'$ and $+$ is faithful. $\square$

Before moving on with the rest of the properties we need to mention a very useful characterisation of extensivity.

Theorem 3.3.1. *A category with sums $(\mathcal{E}, +)$ is extensive if and only if it has pullbacks along coproduct injections and every commutative diagram*

$$\begin{array}{ccccc}
A & \xrightarrow{\alpha} & C & \xleftarrow{\beta} & B \\
f_A \downarrow & & \downarrow f & & \downarrow f_B \\
X & \xrightarrow{\iota_X} & X+Y & \xleftarrow{\iota_Y} & Y
\end{array}$$

comprises a pair of pullback squares if and only if the top row is a sum.

Proof. We first begin by assuming $\mathcal{E}$ extensive and considering a commutative diagram as above. Since we are in an extensive category, we know that we can write the middle arrow as a sum

$$\begin{array}{ccccc}
A & \xrightarrow{\alpha} & A' + B' & \xleftarrow{\beta} & B \\
f_A \downarrow & & \downarrow f_{A'} + f_{B'} & & \downarrow f_B \\
X & \xrightarrow{\quad} & X + Y & \xleftarrow{\quad} & Y
\end{array}
\quad \text{with} \quad
\begin{array}{cc}
A' & B' \\
\downarrow f_{A'} & \downarrow f_{B'} \\
X & Y
\end{array}$$

Suppose first that this diagram is made up of two pullback squares. Then

we get immediately an arrow $a' : A' \rightarrow A$ by

$$\begin{array}{ccccc}
 A' & & \xrightarrow{\iota_{A'}} & & A' + B' \\
 \searrow \exists! a' & & \downarrow \alpha & & \downarrow f_{A'} + f_{B'} \\
 A & \xrightarrow{\alpha} & A' + B' & & \\
 \downarrow f_A & \lrcorner & \downarrow f_{A'} + f_{B'} & & \\
 X & \xrightarrow{\quad} & X + Y & &
 \end{array}$$

such that

$$f_{A'} = f_A \circ a' \quad (2)$$

$$\iota_{A'} = \alpha \circ a' \quad (3)$$

and analogously for b' . We now construct an arrow in the opposite direction. By the property of sums we have a unique arrow induced by α and β :

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} : A + B \longrightarrow A' + B'$$

Now, since the diagram

$$\begin{array}{ccc}
 A + B & \xrightarrow{\begin{pmatrix} \alpha \\ \beta \end{pmatrix}} & A' + B' \\
 \searrow f_A + f_B & & \swarrow f_{A'} + f_{B'} \\
 & X + Y &
 \end{array}$$

commutes, by extensivity we can write $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ as a sum $a + b$ with $a : A \rightarrow A'$ and $b : B \rightarrow B'$. We claim that these are inverses to the a' and b' given by the pullbacks.

Firstly, notice that, since

$$(f_{A'} + f_{B'}) \circ (a + b) = f_A + f_B$$

we have $f_A = f_{A'} \circ a$. By combining this with (2) we get

$$f_A = f_A \circ a' \circ a$$

Furthermore, since $\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = a + b$, we also have $\alpha = \iota_{A'} \circ a$. By combining this with (3) we get

$$\alpha = \alpha \circ a' \circ a$$

Thus the map $a' \circ a$ acts on f_A and α just like id_A . By the property of pullbacks

$$\begin{array}{ccc}
 A & \xrightarrow{\alpha} & A' + B' \\
 \downarrow f_A & \lrcorner & \downarrow f_{A'} + f_{B'} \\
 X & \xrightarrow{\quad} & X + Y
 \end{array}$$

we get

$$a' \circ a = id_A$$

To check the other identity, consider $aa' + bb' : A' + B' \rightarrow A' + B'$

$$\begin{array}{ccc}
 A' & \xrightarrow{\iota_{A'}} & A' + B' \\
 \downarrow aa' & \searrow \iota_{A'} & \downarrow aa' + bb' \\
 A' & \xrightarrow{\iota_{A'}} & A' + B'
 \end{array}$$

Clearly

$$(aa' + bb') \circ \iota_{A'} = \iota_{A'} \circ a \circ a' = \alpha \circ a' = \iota_{A'}$$

and analogously for B' . Thus $aa' + bb'$ is the sum of two arrows that behaves exactly like $id_{A'} + id_{B'}$. By the faithfulness of $+$ we get

$$aa' = id_{A'} \quad \text{and} \quad bb' = id_{B'}$$

hence proving that the top row is indeed (isomorphic to) a sum.

Conversely, consider a square composed of two sums

$$\begin{array}{ccccc}
 A & \xrightarrow{\iota_A} & A + B & \xleftarrow{\iota_B} & B \\
 \downarrow f_A & & \downarrow f_A + f_B & & \downarrow f_B \\
 X & \xrightarrow{\iota_X} & X + Y & \xleftarrow{\iota_Y} & Y
 \end{array}$$

To see that the square on the left is a pullback, consider arrows $\alpha' : A' \rightarrow A + B$ and $f_{A'} : A' \rightarrow X$ such that

$$(f_A + f_B) \circ \alpha' = \iota_X \circ f_{A'}$$

Now, consider the arrow given by the universal property of the sum

$$\begin{array}{ccccc}
 A' & \xrightarrow{\quad} & A' + B & \xleftarrow{\quad} & B \\
 & \searrow \alpha' & \downarrow (\alpha'_{\iota_B}) & \swarrow \iota_B & \parallel \\
 & A & \xrightarrow{\quad} & A + B & \xleftarrow{\quad} B \\
 & \downarrow & & \downarrow & \downarrow \\
 & X & \xrightarrow{\quad \iota_{A'} \quad} & X + Y & \xleftarrow{\quad} Y
 \end{array}$$

(Note: In the original image, there is a curved arrow labeled $f_{A'}$ from A' to X .)

Since $+$ is fully faithful, the arrow (α'_{ι_B}) can be written uniquely as

$$\begin{pmatrix} \alpha' \\ \iota_B \end{pmatrix} = a' + id_B$$

Thus, in particular, there exists only one $a' : A' \rightarrow A$ such that $\alpha' = \iota_A \circ a'$ and $f_{A'} = f_A \circ a'$, hence the square on the left (and analogously that on the right) is a pullback.

This also tells us that in $\mathcal{E}$ pullbacks of coproduct injections always exist. Indeed, to pull back an arrow $f : C \rightarrow X + Y$ along ι_X we only need to notice that, by extensivity, the arrow f splits in $f_A + f_B : A + B \rightarrow X + Y$. Then we can simply construct the square

$$\begin{array}{ccc}
 A & \xrightarrow{\iota_A} & A + B \\
 f_A \downarrow & & \downarrow f_A + f_B \\
 X & \xrightarrow{\iota_X} & X + Y
 \end{array}$$

And this is a pullback by what we just proved.

Now, to prove the converse, assume that diagram of the form above are sums if and only if they are made of pullbacks, and that pullbacks of injections⁸ exist.

Essential Surjectivity Consider an arrow $f : C \rightarrow X + Y$ and construct the pullbacks along the injections ι_X and ι_Y

$$\begin{array}{ccccc}
 A & \xrightarrow{\quad} & C & \xleftarrow{\quad} & B \\
 \downarrow & \lrcorner & \downarrow f & \lrcorner & \downarrow \\
 X & \xrightarrow{\quad \iota_X \quad} & X + Y & \xleftarrow{\quad \iota_Y \quad} & Y
 \end{array}$$

⁸From hereon we will often write “injections” to mean “coproduct injections”.

By hypothesis, the top row is a sum and thus

$$\begin{array}{ccc}
 C & & A + B \\
 \downarrow f & = & \downarrow f_A + f_B \\
 X + Y & & X + Y
 \end{array}$$

Fullness Consider the following commutative diagram

$$\begin{array}{ccc}
 A + B & \xrightarrow{\varphi} & C + D \\
 \searrow f_A + f_B & & \swarrow g_C + g_D \\
 & X + Y &
 \end{array}$$

Since the square with top row $C \rightarrow C + D \leftarrow D$ is made of pullbacks, we can construct two arrows φ_A and φ_B in the following way

$$\begin{array}{ccccc}
 A & & & & B \\
 \searrow \exists! \varphi_A & \swarrow \iota_A \circ \varphi & & \nwarrow \iota_B \circ \varphi & \swarrow \exists! \varphi_B \\
 C & \xrightarrow{\quad} & C + D & \xleftarrow{\quad} & D \\
 \downarrow f_A & \lrcorner & \downarrow g_C + g_D & \lrcorner & \downarrow f_B \\
 X & \xrightarrow{\iota_X} & X + Y & \xleftarrow{\iota_Y} & Y
 \end{array}$$

Thus $\varphi = \varphi_A + \varphi_B$ and $+$ is full.

□

We claimed that the reason why $\mathcal{P}(X)$ failed to be extensive was that its sums were not *disjoint*. To define this property more precisely:

Definition 3.3.1. Let $\mathcal{C}$ be a category with sums and pullbacks along coproduct injections. We say that sums are **disjoint** in $\mathcal{C}$ if

- i. Pulling back an injection of a binary sum along the other injection yields the initial object.
- ii. All injections are monic.

Proposition 3.3.1. In an extensive category $\mathcal{E}$, sums are disjoint.

Proof. Consider the following diagram

$$\begin{array}{ccccc}
 0 & \xrightarrow{!_Y} & Y & \xlongequal{\quad} & Y \\
 \downarrow !_X & & \downarrow \iota_Y & & \parallel \\
 X & \xrightarrow{\iota_X} & X + Y & \xleftarrow{\iota_Y} & Y
 \end{array}$$

Since the top row is a sum, the squares are both pullbacks. The square on the left being a pullback implies i_* , while the square on the right implies ii . $\square$

Example 3.3.1. Sums in $\mathcal{P}(X)$ are *not* disjoint. Indeed, take two sets A and B . The pullback of the injections - which is also equal for every other C containing $A \cup B$ - is $A \cap B$, which need not to be 0 for every A, B .

3.4 The Burnside rig of a preextensive category

We would like to bring the reader's attention to the proof of Proposition 2.5.1. The main argument was to associate a polynomial to every polyhedral sets and then the thesis followed from a simple polynomial identity. This approach to treat objects in a category as elements of a ring cannot be applied to every category, since there is no guarantee that isomorphism classes of objects in a category - without any additional assumption - will satisfy the axioms of a ring. In this subsection we'll see that a preextensive category will come as close as possible to behaving like a ring in this sense.

We already mentioned the intuition according to which extensivity means that sums are well behaved. Since the initial object 0 is itself a sum, it only makes sense that it has some nice property too.

Definition 3.4.1. An initial object is said to be **strict** if any arrow into it is invertible.

Proposition 3.4.1. In an extensive category $\mathcal{E}$, initials are strict.

Proof. Let $f : A \rightarrow 0$. The fact that $f \circ !_A = id_0$ is true in every category, so let us check the other composition. Consider the diagram

$$\begin{array}{ccccc}
 A & \xlongequal{\quad} & A & \xleftarrow{!_A} & 0 \\
 \downarrow f & & \downarrow f & & \parallel \\
 0 & \xlongequal{\iota_X} & 0 & \xlongequal{\iota_Y} & 0
 \end{array}$$

Since both rows are sums, the squares are pullbacks. In particular, the right square being a pullback implies that

$$\begin{array}{ccccc}
 & & & & A \\
 & & & \swarrow f & \\
 A & \xleftarrow{!_A} & 0 & \xleftarrow{f} & A \\
 \downarrow f & & \downarrow & & \downarrow f \\
 0 & \xlongequal{\iota_Y} & 0 & & 0
 \end{array}$$

Hence $!_A \circ f = id_A$. $\square$

Corollary 3.4.1. *In an extensive category $\mathcal{E}$ with products, we have*

$$A \times 0 = 0$$

for each object A .

Proof. Since initials are strict, the projection $\pi_0 : A \times 0 \rightarrow 0$ is an isomorphism. $\square$

This unassuming corollary is the first step towards our program of treating objects in a preextensive category as polynomials. The second step is given by the following proposition.

Proposition 3.4.2. *If $\mathcal{E}$ is an extensive category with finite products, then $\mathcal{E}$ is also distributive.*

Proof. Take objects A, B, C , and consider the diagram given by

$$\begin{array}{ccccc} A \times B & \xrightarrow{id_A \times \iota_B} & A \times (B + C) & \xleftarrow{id_A \times \iota_C} & A \times C \\ \pi_B \downarrow & & \pi_{B+C} \downarrow & & \downarrow \pi_C \\ B & \xrightarrow{\iota_B} & B + C & \xleftarrow{\iota_C} & C \end{array}$$

It is clear that the thesis consists in proving that the top row is a sum. Being in an extensive category, this reduces to proving that the squares are pullbacks. We take for example the square on the left and build a cone over it.

$$\begin{array}{c} D \xrightarrow{\quad g \quad} A \times (B + C) \\ \quad \searrow f \quad \downarrow \pi_{B+C} \\ \quad \quad A \times B \xrightarrow{id_A \times \iota_B} B + C \\ \quad \quad \downarrow \pi_B \quad \downarrow \iota_B \\ \quad \quad B \end{array}$$

To build our wanted arrow from D to $A \times B$ we simply consider the arrow $\langle f, g' \rangle$ where $g' = \pi_A \circ g$. We need to check that

$$g = (id_A \times \iota_B) \circ \langle f, g' \rangle$$

But this is given by the following commutative diagram.

$$\begin{array}{ccccc}
& & D & \xrightarrow{g} & A \times (B + C) \\
& \swarrow f & \downarrow \langle f, g' \rangle & \searrow g' & \downarrow \pi_A \\
B & \xleftarrow{\pi_B} & A \times B & \xrightarrow{\pi_A} & A \\
\downarrow \iota_B & & \downarrow id_A \times \iota_B & & \parallel \\
B + C & \xleftarrow{\pi_{B+C}} & A \times (B + C) & \xrightarrow{\pi_A} & A
\end{array}$$

In particular, the path on the right tells us that

$$\pi_A \circ g = \pi_A \circ (id_A \times \iota_B) \circ \langle f, g' \rangle$$

While the path on the left tells us that

$$\begin{aligned}
\pi_{B+C} \circ (id_A \times \iota_B) \circ \langle f, g' \rangle &= \iota_B \circ \pi_B \circ \langle f, g' \rangle \\
&= \iota_B \circ f \\
&= \pi_{B+C} \circ g
\end{aligned}$$

So the obvious arrow works. Is it unique? Let φ be another arrow such that

$$\pi_B \circ \varphi = f \quad \text{and} \quad (id_A \times \iota_B) \circ \varphi = g$$

We only need to post-compose φ with π_A and π_B to check that it is equal to $\langle f, g' \rangle$. One equality, $\pi_B \varphi = f$ is given by the very definition of φ . As for the other one,

$$g' = \pi_A g = \pi_A (id_A \times \iota_B) \varphi = id_A \pi_A \varphi = \pi_A \varphi$$

Thus the arrow $\langle f, g' \rangle$ is unique, the squares are pullbacks, and our category is distributive. $\square$

Any category satisfying the above properties comes equipped with a canonical associated algebraic structure, called its *Burnside rig*⁹.

Definition 3.4.2. Let $\mathcal{E}$ be a category with finite sums and products satisfying

$$A \times 0 \simeq 0 \quad (\text{Absorption Law})$$

$$A \times (B + C) \simeq A \times B + A \times C \quad (\text{Distributivity Law})$$

for each object A, B and C . We define its **Burnside rig** (denoted with $\mathcal{B}(\mathcal{E})$) as

$$\mathcal{B}(\mathcal{E}) = (Ob(\mathcal{E}) / \simeq, +, \times)$$

⁹The terminology comes from Lawvere and Schanuel.

Notice how $\mathcal{B}(\mathcal{E})$ is not a ring, but a rig: a poorer and more complicated algebraic structure to study.

Example 3.4.1. Let $\mathcal{E}$ be $\mathbf{Set}_f$, the category of finite sets. Then its Burnside rig $\mathcal{B}(\mathcal{E})$ is the set of natural numbers $\mathbb{N}$.

Definition 3.4.3. A rig is an algebraic structure consisting of a set R an addition $+$ and a multiplication $\cdot$ such that

- Both $(R, +)$ and $(R, \cdot)$ are commutative¹⁰ monoids.
- The multiplication distributes over addition, i.e. it holds

$$t \cdot (r + s) = t \cdot r + r \cdot s$$

- The absorption law

$$r \cdot 0 = 0$$

holds.

The intuition behind this definition is that a rig is just a ring without the assumption that every element has an additive inverse. One can think of a rig as a ring without negatives, hence dropping the “n”.

Example 3.4.2. • Every (commutative) ring is also a rig.

- The set of natural numbers $\mathbb{N}$ is a rig.
- Every distributive lattice is a rig.
- Given a rig R , the set of polynomials with coefficients in R is a rig.
- Every rig with positive characteristic is a ring.

Remark 3.4.1. Why a rig? Wouldn’t it be simpler to study the categories whose objects behave like elements of a ring? Certainly. The only problem is that such a category does not exist.

Indeed, if $\mathcal{C}$ is a category with finite sums and A, B are objects such that $A + B = 0$ then both A and B are 0. Indeed, let X be any other object. Since $A + B = 0$ there will exist a (unique) arrow from $A + B$ to X . By pre-composing it with ι_A we get

$$A \xrightarrow{\iota_A} A + B = 0 \xrightarrow{!_X} X$$

¹⁰Just like rings, one can make a distinction between commutative and non-commutative rigs. Since our rigs originate from a commutative structure, we will omit the “commutative” term for brevity’s sake.

So an arrow from A to any object X always exists. Moreover, if $f : A \rightarrow X$ is another arrow, by the universal property of the sum

$$\begin{array}{ccccc}
 A & \xrightarrow{\iota_A} & A + B & \xleftarrow{\iota_B} & B \\
 & \searrow f & \downarrow \exists! g & \swarrow !_X \circ \iota_B & \\
 & & X & &
 \end{array}$$

we get a unique $g : A + B \rightarrow X$ such that $f = g \circ \iota_A$. But since $A + B = 0$, there is only one arrow from $A + B$ in X and that is precisely $!_X$. This means that $f = !_X \circ \iota_A$ and thus $A = 0$.

Remark 3.4.2. One could object that, while it's true that a category with negative objects cannot exist, we can always “complete” a Burnside rig to a whole ring by virtually adding negatives. After all, negative numbers do not exist in the “real” world, but this cannot stop us from studying them. The way to do that is simple: the forgetful functor from the category of rings to that of rigs admits a left adjoint¹¹ Gr. Note that this is the approach typically used in the field of K -theory.

3.5 Partitions in an extensive setting

Probably the most important property of the Euler characteristic is that it does not depend on the decomposition of polyhedral sets. Meaning that if the same polyhedral set P can be written as a sum of polyhedral sets in two different ways

$$\sum_{i \in I} A_i = P = \sum_{j \in J} B_j$$

the Euler characteristic is unchanged.

$$\sum_{i \in I} \chi(A_i) = \sum_{j \in J} \chi(B_j)$$

In order to generalize this notion to extensive categories one has to talk of joint refinements of partitions, that is a collection C_{ij} of objects such that

$$\sum_{i \in I} C_{ij} = B_j \quad \text{and} \quad \sum_{j \in J} C_{ij} = A_i$$

We prove that, under an additional hypothesis, partitions in extensive categories admit a joint refinement. Please note that the following definitions and concepts are just a proposal from the author and are in no way standard.

¹¹Sometimes called “group completion” or “Groethendieck completion”.

Definition 3.5.1. Let $\mathcal{C}$ be a category with sums. We say that a (finite) partition of an object C is a (finite) collection of non-initial subobjects $a_i : A_i \rightarrow C$ (for $i \in I$) such that

$$(a_1, \dots, a_I) : \sum_i A_i \longrightarrow C$$

is an isomorphism.

Note that generally, to speak about partitions one needs to have a commutative monoid $(M, +)$ with the property

$$A + B = 0 \implies A = B = 0$$

Remark 3.4.1 guarantees that the set $(Ob(\mathcal{C})/\simeq, +)$ is such a monoid.

Remark 3.5.1. Note that, generally speaking, summands are not necessarily subobjects of the sum since injections need not to be monomorphisms. The perk of working in an extensive setting is that Proposition 3.3.1 guarantees us that injections are in fact monic.

Definition 3.5.2. Let $\mathcal{C}$ be a category with sums and a terminal object. A **partition classifier** is a collection of arrows $t_i : 1 \rightarrow \Omega_I$ (for $i \in I$) such that for every partition $\{a_i\}$ of an object C there exists a unique arrow $a : C \rightarrow \Omega_I$ such that the following diagram

$$\begin{array}{ccc} A_i & \xrightarrow{!_{A_i}} & 1 \\ a_i \downarrow & & \downarrow t_i \\ C & \xrightarrow{a} & \Omega_I \end{array}$$

is a pullback, for each $i \in I$.

This definition should remind readers of a subobject classifier and they could be wondering if one implies the other. The answer is yes in a nice enough category.

Definition 3.5.3. An extensive category with terminal objects is called **Boolean** if the first injection

$$t : 1 \longrightarrow 1 + 1$$

is a subobject classifier.

Example 3.5.1. Set is a Boolean category.

Proposition 3.5.1. If $\mathcal{E}$ is Boolean then it admits a partition classifier. Moreover, the partition classifier in this case is the collection

$$\iota_i : 1 \longrightarrow I = \underbrace{1 + \dots + 1}_{I \text{ times}}$$

Proof. Let us reason by induction on I .

Base case $I = 2$ Take a partition of just two elements $A_1 + A_2 = C$ and consider the pullback square given by the subobject classifier of A_1 :

$$\begin{array}{ccc} A_1 & \xrightarrow{!_1} & 1 \\ \downarrow a_1 & \lrcorner & \downarrow \iota_1 \\ C & \xrightarrow{\chi_1} & 1 + 1 \end{array}$$

We can complete this square with its ‘companion’ in the following way:

$$\begin{array}{ccccc} A_1 & \xrightarrow{a_1} & C & \xleftarrow{a_2} & A_2 \\ \downarrow !_{A_1} & \lrcorner & \downarrow \chi_1 & & \downarrow !_{A_2} \\ 1 & \xrightarrow{\iota_1} & 1 + 1 & \xleftarrow{\iota_2} & 1 \end{array}$$

Since $\mathcal{E}$ is extensive and the top row is a sum, the right square is also a pullback, which is what we needed to get a partition classifier.

Inductive case Take a partition of I elements, $C = A_1 + A_2 + \dots + A_I$ and consider the binary partition $C = A + B$ given by

$$A = A_1 \quad B = \underbrace{A_2 + \dots + A_I}_{I-1 \text{ terms}}$$

By inductive hypothesis we have a partition classifier for B and thus the following pullback squares

$$\begin{array}{ccc} A_i & \xrightarrow{!_1} & 1 \\ \downarrow a'_i & \lrcorner & \downarrow \iota_1 \\ B & \xrightarrow{b} & I - 1 \end{array}$$

for each $i > 1$. Consider now the following diagram

$$\begin{array}{ccccc} A & \xrightarrow{a_1} & C & \xleftarrow{(a_2, \dots, a_I)} & B \\ \downarrow !_A & & \downarrow !_{A+b} & & \downarrow b \\ 1 & \xrightarrow{\iota_1} & I & \xleftarrow{\iota_{I-1}} & I - 1 \end{array}$$

Since the top row is a sum, both squares are pullbacks. Moreover, for each $i > 1$ we can compose one of the previous squares on the right:

$$\begin{array}{ccccc}
C & \xleftarrow{(a_2, \dots, a_I)} & B & \xleftarrow{a'_i} & A_i \\
\downarrow !_{A+b} & \lrcorner & \downarrow b & \lrcorner & \downarrow !_i \\
I & \xleftarrow{\iota_{I-1}} & I-1 & \xleftarrow{\iota_i} & 1
\end{array}$$

Thus the composition is still a pullback and $\iota_i : 1 \rightarrow I$ is a partition classifier.

□

I am currently not aware of any partition classifiers other than the one above.

Remark 3.5.2. A reasonable question would be to wonder if some sort of converse holds. The answer is no and the reason is simple: in general, partitionable objects are way scarcer than objects that admit subobjects. For example, take a sphere S^2 in the category of smooth manifolds. Clearly S^2 has many non-trivial subobjects (e.g. its equator S^1) but it cannot be partitioned in a sum since it's connected. Thus, even if a category has the means to classify every single partition, it cannot hope to classify every single subobject. That being said, we can aim lower to achieve a partial result.

Definition 3.5.4. Let C be a class of monomorphisms in $\mathcal{E}$. A C -subobject classifier is a C -monomorphism $t : 1 \rightarrow \Omega$ out of the terminal object, such that for every C -monomorphism $U \rightarrow X$ in $\mathcal{E}$ there is a unique morphism $\chi : X \rightarrow \Omega$ such that there is a pullback diagram of the following form:

$$\begin{array}{ccc}
U & \xrightarrow{!_U} & 1 \\
\downarrow & \lrcorner & \downarrow t \\
X & \xrightarrow{\chi} & \Omega
\end{array}$$

Proposition 3.5.2. If $\mathcal{E}$ is an extensive category with a terminal object, then $\mathcal{E}$ is **weakly Boolean** in the following sense. The inclusion

$$\iota_1 : 1 \longrightarrow 1 + 1$$

is an S -subobject classifier where S is the class of monomorphisms that are inclusions in a sum.

Proof. Take an S -monomorphism, i.e. an inclusion in a sum $A \hookrightarrow A + B$, and

consider the usual diagram

$$\begin{array}{ccccc}
A & \xrightarrow{\iota_A} & A + B & \xleftarrow{\iota_B} & B \\
\downarrow !_A & \lrcorner & \downarrow !_A + !_B & \lrcorner & \downarrow !_B \\
1 & \xrightarrow{\iota_1} & 1 + 1 & \xleftarrow{\iota_2} & 1
\end{array}$$

We know that the squares are pullbacks. To check uniqueness, consider another arrow $\chi : A + B \rightarrow 1 + 1$ such that

$$\begin{array}{ccc}
A & \xrightarrow{\iota_A} & A + B \\
\downarrow !_A & \lrcorner & \downarrow \chi \\
1 & \xrightarrow{\iota_1} & 1 + 1
\end{array}$$

is a pullback. By extensivity, χ must split in the sum of two arrows and those arrows are necessarily $!_A + !_B$, proving the thesis. $\square$

Example 3.5.2. One could think that the Boolean definition might be too strong and could be weakened. After all, if we are in presence of terminals, extensivity and a subobject classifier, it seems only natural that the subobject classifier must be $1 + 1$. Unfortunately, this is not the case. Take for example the category of functions $\mathcal{E} = \mathbf{Set}^{0 \rightarrow 1}$, whose objects are functions and whose arrows are commutative diagram. This is extensive and admits terminals and subobject classifier, but its subobject classifier is the arrow

$$\begin{aligned}
\omega : \left\{ 0, \frac{1}{2}, 1 \right\} &\longrightarrow \{0, 1\} \\
0 &\longmapsto 0 \\
\frac{1}{2}, 1 &\longmapsto 1
\end{aligned}$$

which is very different from the arrow $1 + 1 = id_{\{0,1\}}$.

Thus Booleanity is a very strong property that not many categories possess. This in turn gives us some particularly good results.

Proposition 3.5.3. *Let $\mathcal{E}$ be a Boolean category. Then different partitions of the same object admit a common refinement, meaning that if $\sum_i A_i = \sum_j B_j$ then there exist a collection of objects $\{C_{ij}\}$ such that*

$$\sum_j C_{ij} = A_i \quad \text{and} \quad \sum_i C_{ij} = B_j$$

Proof. Let's call C the partitioned object $C = \sum_i A_i = \sum_j B_j$. Since $\mathcal{E}$ is Boolean, by Proposition 3.5.1 we can classify our two partitions with two arrows

$$C \xrightarrow{a} I \qquad C \xrightarrow{b} J$$

such that, for every i and j , the following diagrams

$$\begin{array}{ccc} A_i & \xrightarrow{!_{A_i}} & 1 \\ a_i \downarrow & & \downarrow \iota_i \\ C & \xrightarrow{a} & I \end{array} \qquad \begin{array}{ccc} B_j & \xrightarrow{!_{B_j}} & 1 \\ b_j \downarrow & & \downarrow \iota_j \\ C & \xrightarrow{b} & J \end{array}$$

are pullbacks. Now, since our goal is to find a (common) partition of A_i and B_j , we will need arrows

$$A_i \xrightarrow{f_i} J \qquad B_j \xrightarrow{g_j} I$$

Consider now

$$f_i = b \circ a_i \quad \text{and} \quad g_j = a \circ b_j$$

Now, since injections can always be pulled back in an extensive category, we can form the following squares

$$\begin{array}{ccc} C_{ij} & \xrightarrow{!} & 1 \\ c_{ij} \downarrow & \lrcorner & \downarrow \iota_i \\ B_j & \xrightarrow{g_j} & I \end{array} \qquad \begin{array}{ccc} C'_{ij} & \xrightarrow{!} & 1 \\ c'_{ij} \downarrow & \lrcorner & \downarrow \iota_j \\ A_i & \xrightarrow{f_i} & J \end{array}$$

Our efforts will now be devoted to proving that $C_{ij} \simeq C'_{ij}$. Consider the following diagram

$$\begin{array}{ccccc} C'_{ij} & & & & \\ \downarrow c'_{ij} & \searrow !'_{ij} & & & \\ & & B_j & \xrightarrow{!_j} & 1 \\ & & \downarrow b_j & \lrcorner & \downarrow \iota_j \\ & & A_i & & \\ & \searrow a_i & & & \\ & & C & \xrightarrow{b} & J \end{array}$$

Note that

$$b \circ a_i \circ c'_{ij} = f_i \circ c'_{ij} = \iota_j \circ !'_{ij}$$

Thus, by the pullback property, we get a unique arrow $\sigma_{ij} : C'_{ij} \rightarrow B_j$, and analogously a unique arrow $\tau_{ij} : C_{ij} \rightarrow A_i$, such that

$$b_j \circ \sigma_{ij} = a_i \circ c'_{ij} \quad \text{and} \quad a_i \circ \tau_{ij} = b_j \circ c_{ij}$$

Consider now the following square

$$\begin{array}{ccccc}
 C'_{ij} & & & & \\
 \downarrow \sigma_{ij} & \searrow \text{!}_{ij}' & & & \\
 & C_{ij} & \xrightarrow{\text{!}_{ij}} & 1 & \\
 & \downarrow c_{ij} & \lrcorner & \downarrow \iota_i & \\
 & B_j & \xrightarrow{g_j} & I &
 \end{array}$$

Observe that

$$\begin{aligned}
 g_j \circ \sigma_{ij} &= a \circ b_j \circ \sigma_{ij} \\
 &= a \circ a_i \circ c'_{ij} \\
 &= \iota_i \circ \text{!}_{A_i} \circ c'_{ij} \\
 &= \iota_i \circ \text{!}_{ij}'
 \end{aligned}$$

Hence, by the pullback property, we obtain a unique arrow $\varphi : C'_{ij} \rightarrow C_{ij}$, and analogously a unique arrow $\psi : C_{ij} \rightarrow C'_{ij}$ such that

$$c_{ij} \circ \varphi = \sigma_{ij} \quad \text{and} \quad c'_{ij} \circ \psi = \tau_{ij}$$

We claim that φ and ψ are inverses of each other. To see this, take the equation $c_{ij} \circ \varphi = \sigma_{ij}$, pre-compose ψ and post-compose b_j :

$$\begin{array}{ccc}
 b_j c_{ij} \varphi \psi & \equiv & b_j \sigma_{ij} \psi \\
 \parallel & & \parallel \\
 a_i \tau_{ij} \varphi \psi & \equiv & a_i c'_{ij} \psi
 \end{array}$$

Now, since a_i is monic, we can cancel it

$$\tau_{ij} \varphi \psi = c'_{ij} \psi = \tau_{ij}$$

Thus, if τ_{ij} were monic, we would get the desired equality $\varphi \psi = id_{C_{ij}}$. To see that this is in fact the case, notice how

$$a_i \circ \tau_{ij} = b_j \circ c_{ij}$$

is monic since it's a composition of monos. Moreover, if $a_i \circ \tau_{ij}$ is monic, τ_{ij} must be monic as well. Thus

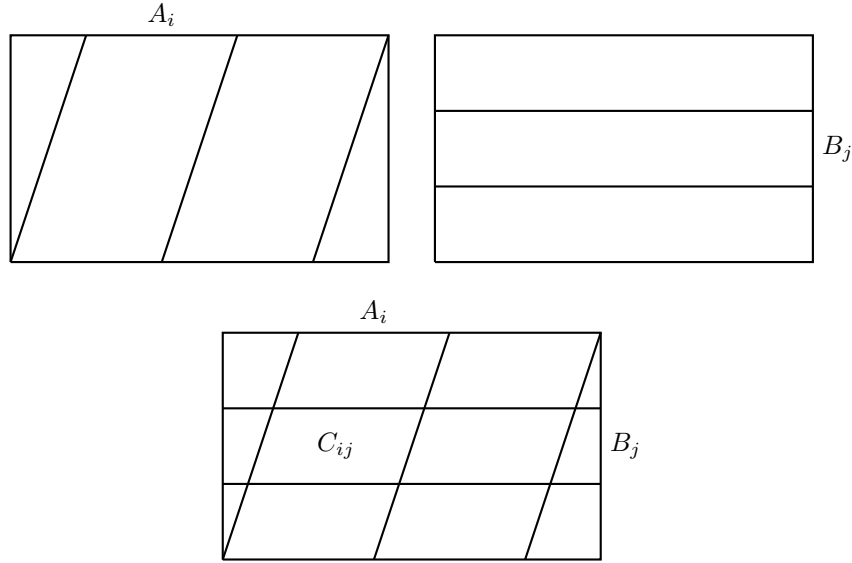
$$\varphi \psi = id_{C_{ij}}$$

and analogously

$$\psi\varphi = id_{C'_{ij}}$$

ending our proof. $\square$

Remark 3.5.3. Intuitively speaking, the partition f_i is obtained by restricting the partition b to one single subobject A_i , identified by a_i . Thus the proposition tells us that, in a Boolean category, partitions are inherited by subobjects, similarly to what happens in **Set**.



Remark 3.5.4. Given an object C , refinements induce a partial ordering on the set $\text{Part}(C)$ by defining

$$c \leq a \iff c \text{ is a refinement of } a \iff \exists h \text{ s.t. } a = h \circ c$$

Thus Proposition 3.5.3 basically tells us that such a poset admits a “meet” operation $\wedge$.

4 Schanuel's Theorem for Polyhedral Sets

4.1 Goals and acknowledgments

We know the Euler-Poincaré characteristic is a topological invariant, meaning that two homeomorphic smooth manifolds will have the same characteristic. So it is only natural to expect a similar result to hold for the polyhedral Euler characteristic, too. The question we wish to answer - though ill-posed as yet - is the following:

Question 1

How much does the Euler characteristic discerns polyhedral sets?

For example, a closed line segment and a point will have the same Euler characteristic $\chi = 1$ but they are clearly different, since they have different dimensions. So what about polyhedra with the same Euler characteristic and dimension? A remarkable corollary of a Theorem due to Schanuel [14] states that the dimension and the Euler characteristic is actually all that matters.

Theorem 4.1.1 (Main Theorem). *Let P and Q be two polyhedral sets. Then $P \simeq Q$ if and only if*

$$\dim(P) = \dim(Q) \quad \text{and} \quad \chi(P) = \chi(Q)$$

Our efforts in this chapter will be devoted to proving Schanuel's Theorem and its remarkable corollary.

This is a truly beautiful and clean theorem, one that says that a simple act of counting encapsulates each and every geometric information about a polyhedron. The only problem of this theorem is that it's also categorically false.

Example 4.1.1. Consider the following polyhedral sets:

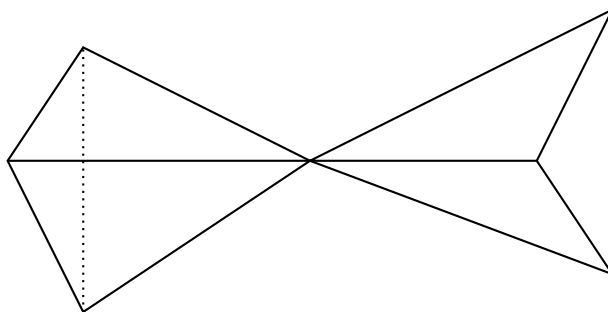


Fig.1: P

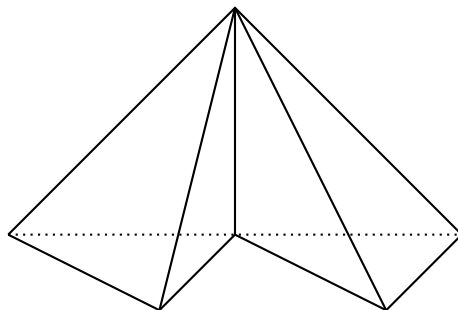


Fig.2: Q

Now, it's easy to see that $\dim(P) = \dim(Q)$ and moreover that $\chi(P) = \chi(Q) = 3$, but P and Q can't hope to be homeomorphic. Indeed, by removing from P the point in the middle of the two tetrahedra, we get two different connected components while we have no chances to do the same with any point of Q .

So, our theorem is false. What do we do now? Naturally, we change our hypothesis until the thesis holds. One way would be to limit ourselves to consider only a particular class of polyhedra, those possessing a particular property that would be the polyhedra in Example 4.1.1. Note that this type of “monster-barring” approach is exactly the one Ernest de Jonquières pursued in his talks against the detractors of Euler’s Theorem 2.1.1 to the French Academy in 1890. We will take a different road instead; one in which both polyhedra from Example 4.1.1 are not only accepted, but isomorphic too. The trick is to very accurately define morphisms of polyhedra without defaulting to continuous maps.

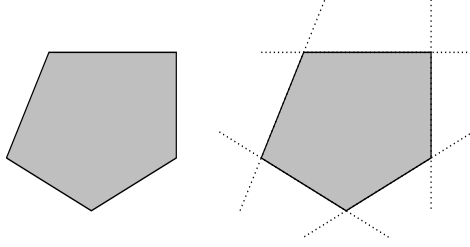
4.2 A category of polyhedral sets

The maps we will consider for our category are functions that will send polyhedral sets to polyhedral sets and with the property that preimages of polyhedral sets are again polyhedral sets.

The intuition behind the choice of maps comes from a characterization of polytopes, the building blocks of our polyhedral sets. We defined them as convex hull of a finite number of points but another way to do that is to consider the intersection of half-spaces given by hyperplanes. Namely, every hyperplane $\mathcal{H} = \{x \in \mathbb{R}^n \mid a^t x = b\}$ determines two halfspaces

$$\mathcal{H}^+ = \{x \in \mathbb{R}^n \mid a^t x \geq b\} \quad \text{and} \quad \mathcal{H}^- = \{x \in \mathbb{R}^n \mid a^t x \leq b\}$$

Any intersection of finitely many half-spaces which is also bounded can be expressed as the convex hull of finitely many points, and the converse is true as well. This result bears the demanding name of Fundamental Theorem of Polytopes and a full account of it can be found in [17].



So the key idea is that a polyhedral map - since it should send a polyhedral set to another - should send a hyperplane into another. The maps that do so are obviously affine maps. But since a polyhedral set is made up of finitely many affine pieces (its faces) then our maps should be made up of finitely many affine maps as well.

Definition 4.2.1. A map f between two polyhedral sets $P \subset \mathbb{R}^n$ and $Q \subset \mathbb{R}^m$ is called **polyhedral** if its graph $\Gamma(f) \subset \mathbb{R}^{n+m}$ is a polyhedral set.

An easy but useful characterization of a polyhedral map is the following.

Lemma 4.2.1. A map $f : P \rightarrow Q$ is polyhedral if and only if it is piecewise-linear.¹² Meaning that there exist two partitions of $P = \bigcup P_i$ and $Q = \bigcup Q_j$ such that for every i the restriction $f \upharpoonright P_i$ is an affine map that sends P_i to some Q_j .

Proof. It's clear that every affine map is polyhedral. Moreover, if f is piecewise-linear with $f_i = f \upharpoonright P_i$ being affine maps, we have

$$\Gamma(f) = \bigcup \Gamma(f_i)$$

and thus f is polyhedral. Conversely, let f be a polyhedral map. Since $\Gamma(f)$ is a polyhedral set, we can write it in disjunctive normal form as

$$\Gamma(f) = \bigvee (T_k - S_k)$$

where T_k and S_k are polyhedra. Now, for each k consider a system of faces $\{G_{hk}\}$ for T_k and let

$$G'_{hk} = G_{hk} - S_k$$

These are a collection of polyhedral sets that covers $\Gamma(f)$, but it fails to be a partition as it is not made of pairwise disjoint sets. To get such a partition, consider the collection $\{G''_{hk}\}$ defined by

$$G''_{hk} = G'_{hk} - \bigvee G'_{lm}$$

¹²This is an especially unfortunate terminology for two different reasons. The first being that such a map is affine on its pieces, and not linear like the name would suggest. Secondly, authors from the field of piecewise-linear topology use the same exact term which can lead to confusion. The main difference between these two types of maps is that we do not require any continuity condition.

where the G'_{lm} are precisely those not properly containing G'_{hk} . The resulting collection of polyhedral sets, which for brevity's sake we will denote only as $\{G_k\}$, is a partition of $\Gamma(f)$, with each of its components sitting entirely in an affine space. By projecting them on P and Q we get two partitions

$$P_i = \pi_1(G_k) \quad \text{and} \quad Q_j = \pi_2(G_k)$$

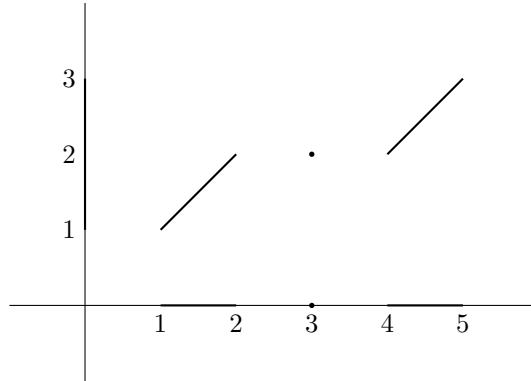
Since each G_k sit in an affine space, $f \upharpoonright P_i$ sends an affine space to another and is thus affine. $\square$

Everything in the remaining pages of this chapter lays on this apparently innocuous definition. For example, by refining the partitions above one can easily see that composition of polyhedral maps is again a polyhedral map. Another thing to notice is the fact that every topological notion is completely lost:

Example 4.2.1. Let $P = (1, 2) \cup \{3\} \cup (4, 5) \subset \mathbb{R}$ and consider $\varphi : P \rightarrow (1, 3)$ given by

$$\varphi(t) = \begin{cases} t & \text{if } t \in (1, 2) \\ 2 & \text{if } t = 3 \\ t - 2 & \text{otherwise} \end{cases}$$

Now, it's easy to see that $\Gamma(\varphi)$ is a polyhedral set which makes φ a polyhedral map.¹³



Thus a polyhedral map can send a connected set onto a non-connected one, and so it's very distant from being continuous. Doing geometry without topology might sound nonsensical at first but it actually helps us avoid topological monsters like devil ladders and Banach-Tarski paradoxes. The attentive reader would have also noticed that connectedness is precisely what made Example 4.1.1 a counterexample, so it's only natural to get rid of it if we want any chance to prove our theorem.

¹³Moreover, since φ is invertible, it is also an isomorphism of polyhedral sets.

We will denote with the letter $\mathcal{P}$ the category whose objects are bounded polyhedral sets and whose arrows are polyhedral maps. Much of our work in this first chapter will consist in proving nice properties for the category $\mathcal{P}$.

Proposition 4.2.1. *The category $\mathcal{P}$ has finite sums.*

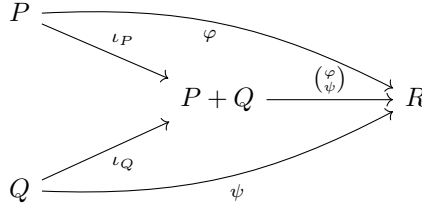
Proof. This proof will be the first example of a fact recurring many times throughout this chapter: $\mathcal{P}$ is really similar to **Set**.

Case 0: $\mathcal{P}$ has initial object Clearly, $\emptyset$ is a finite set and such, by definition, a polyhedron. Furthermore, given P any polyhedral set we can consider the initial map in **Set**: $!_P : \emptyset \rightarrow P$ given by the fact that P is a set. Now, $!_P$ is polyhedral because $\Gamma(!_P) \subset \emptyset \times P = \emptyset$ and thus $\Gamma(!_P)$ is a polyhedral set. Uniqueness is similarly given by the uniqueness in **Set** together with the fact that a polyhedral map has an underlying function.

Case 1: $\mathcal{P}$ has binary sums Once again, the sum in $\mathcal{P}$ is absolutely identical to the one in **Set**: $P + Q = P \sqcup Q$ with the usual injections ι_P, ι_Q . To check that these are indeed polyhedral maps one can note that

$$\Gamma(\iota_P) = \{(p, \iota_P(p)) = (p, p) \mid p \in P\} = \Delta_P \cong P$$

Moreover, the map given by the universal property of sums in **Set** is polyhedral too:



This is because one can easily see that

$$\Gamma\left(\begin{pmatrix} \varphi \\ \psi \end{pmatrix}\right) = \{(p, \varphi(p)) \mid p \in P\} \sqcup \{(q, \psi(q)) \mid q \in Q\} = \Gamma(\varphi) \sqcup \Gamma(\psi)$$

which are, by hypothesis, polyhedral, and so $\begin{pmatrix} \varphi \\ \psi \end{pmatrix}$ is polyhedral too.

□

Proposition 4.2.2. *The category $\mathcal{P}$ has finite limits.*

Proof. **$\mathcal{P}$ has finite products** We already proved in Proposition 2.5.1 that the Cartesian product $P \times Q$ is polyhedral, so we only need to check the

polyhedricity of the product maps:

$$\begin{array}{ccccc}
 & & & & P \\
 & & \nearrow \varphi & & \\
 R & \xrightarrow{\langle \varphi, \psi \rangle} & P \times Q & \xrightarrow{\pi_P} & P \\
 & \searrow \psi & & \searrow \pi_Q & \\
 & & & & Q
 \end{array}$$

One can observe that

$$\Gamma(\pi_P) = \{(p, q, p) \in P \times Q \times P\} \cong \Delta_P \times Q \cong P \times Q$$

and analogously for π_Q . Moreover,

$$\begin{aligned}
 \Gamma(\langle \varphi, \psi \rangle) &= \{(r, \varphi(r), \psi(r)) \mid r \in R\} \cong \{(r, \varphi(r), r, \psi(r)) \mid r \in R\} \\
 &= (\Gamma(\varphi) \times \Gamma(\psi)) \cap (\Delta_R \times P \times Q)
 \end{aligned}$$

This concludes our proof that $\mathcal{P}$ has finite products.

$\mathcal{P}$ has equalizers Let P, Q be two polyhedral sets and φ, ψ be two polyhedral maps between them.

$$\begin{array}{ccc}
 P & \xrightarrow{\varphi} & Q \\
 & \xrightarrow{\psi} &
 \end{array}$$

As before, we will do the obvious thing and prove that the usual equalizer in **Set** works here too. Let $R = \{p \in P \mid \varphi(p) = \psi(p)\}$ and $\sigma : R \rightarrow P$ the obvious inclusion. Clearly,

$$\Gamma(\sigma) = \Gamma(\varphi) \cap \Gamma(\psi)$$

which are polyhedral by hypothesis. Furthermore, since R is the projection of $\Gamma(\sigma)$ on the first $\dim(P)$ coordinates, by the linear case of the Tarski-Seidenberg theorem [12] (or, equivalently, by the quantifier elimination in the theory of real-closed ordered fields) we get that R is polyhedral too. Moreover, if we had another map of polyhedral sets $\tau : F \rightarrow P$ equalizing φ and ψ , the universal map induced by the equalizer property of E in **Set** is just the map τ itself, which is polyhedral by hypothesis. $\square$

So our category has at least *some* good properties. We can also observe another easy fact: if we let Q be a subobject of P , we also have for free that

$$Q^c \cap P$$

is polyhedral too. Thus if $\mathcal{P}$ is extensive, then it is also Boolean since every P can be split in two subobjects.

4.3 The Burnside rig of bounded polyhedral sets

Among the many similarities that $\mathcal{P}$ shares with $\mathbf{Set}$, there is extensivity. This shouldn't surprise anyone, since we've seen how extensivity is a good behaviour of sums and sums in $\mathcal{P}$ are exactly identical to those in $\mathbf{Set}$. Before proving the result, we will state a lemma which is of some help in proving extensivity in a setting with products.

Lemma 4.3.1. *Let $\mathcal{E}$ be a category with finite sums and terminal object. Then $\mathcal{E}$ is extensive if and only if the functor*

$$+ : (\mathcal{E}/1) \times (\mathcal{E}/1) \rightarrow \mathcal{E}/(1+1)$$

is an equivalence.

Proof. Let X and Y be two objects in $\mathcal{E}$ and assume the extensivity condition for 1.

Essential-Surjectivity Let f be an element of $\mathcal{E}/(X+Y)$, i.e. an arrow $C \rightarrow X+Y$. By post-composing it with $!_X+_!_Y$ we get

$$C \xrightarrow{f} X+Y \xrightarrow{!_X+_!_Y} 1+1$$

By the extensivity condition for 1, this arrow splits in $!_A+_!_B : A+B = C \rightarrow 1+1$. Thus we get the following diagram

$$\begin{array}{ccc} C = A+B & \xrightarrow{f} & X+Y \\ & \searrow \scriptstyle !_A+_!_B & \swarrow \scriptstyle !_X+_!_Y \\ & 1+1 & \end{array}$$

By the fullness of $+$ we get that f splits in f_A+f_B , proving the essential surjectivity.

Fullness Consider the following commutative diagram

$$\begin{array}{ccc} A+B & \xrightarrow{\varphi} & C+D \\ & \searrow \scriptstyle f_A+f_B & \swarrow \scriptstyle g_C+g_D \\ & X+Y & \end{array}$$

Clearly, by composing the tip of this diagram with $!_X+_!_Y : X+Y \rightarrow 1+1$ we get the diagram

$$\begin{array}{ccc} A+B & \xrightarrow{\varphi} & C+D \\ & \searrow \scriptstyle !_A+_!_B & \swarrow \scriptstyle !_C+_!_D \\ & 1+1 & \end{array}$$

and, using the fullness of $+$, we can split φ in $\varphi_A+\varphi_B$, ending the proof.

□

Proposition 3.3.1 told us that a category needs disjointness in order to be extensive. Lemma 4.3.1 expands on that, guaranteeing that the extensivity condition needs only to be checked on the quintessential disjoint object, $1 + 1$. If $1 + 1$ fails to be disjoint, everything collapses.

Theorem 4.3.1. *The category of bounded polyhedral sets $\mathcal{P}$ is extensive.*

Proof. Lemma 4.3.1 allows us to check the extensivity condition only on $1 + 1$ so let's do precisely that.

Fullness Let $f : A + B \rightarrow C + D$ be a polyhedral arrow such that the following diagram

$$\begin{array}{ccc} A + B & \xrightarrow{f} & C + D \\ & \searrow \quad \swarrow & \\ & !_A + !_B & !_C + !_D \\ & \searrow \quad \swarrow & \\ & 1 + 1 & \end{array}$$

commutes. We split f by putting

$$f_A = f \restriction_A \quad \text{and} \quad f_B = f \restriction_B$$

Since the diagram above commutes, we know that $f(a) \in C$ for every element $a \in A$ and likewise $f(b) \in D$. Thus $f = f_A + f_B$ as functions in $\mathbf{Set}$. To check polyhedricity, notice how

$$\Gamma(f) \subseteq (A + B) \times (C + D) \quad \Gamma(f_A) \subseteq A \times C \quad \Gamma(f_B) \subseteq B \times D$$

In particular,

$$\Gamma(f_A) = \Gamma(f) \cap (A \times C)$$

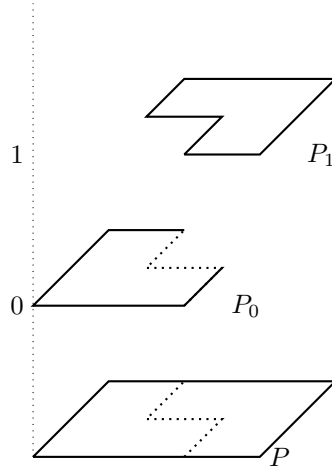
and thus f_A is polyhedral. Analogously for f_B .

Essential Surjectivity Let P be a polyhedral set and $f : P \rightarrow 1 + 1 = \{0, 1\}$ a polyhedral map. We need to split both P and f . Define

$$P_0 = f^{-1}(0) \quad \text{and} \quad P_1 = f^{-1}(1)$$

To show that these are in fact polyhedral, notice that

$$P_i \simeq P_i \times \{i\} \simeq (P \times \{i\}) \cap \Gamma(f)$$



Since both $P \times \{i\}$ and $\Gamma(f)$ are polyhedral, P_i is polyhedral as well. Now, to split f one can just define

$$f_0 = f \upharpoonright_{P_0} \quad f_1 = f \upharpoonright_{P_1}$$

to get $f = f_0 + f_1$. Checking that these maps are polyhedral is identical to the reasoning used for f_A and f_B when checking fullness.

□

Combining this result with Proposition 3.4.2 yields the distributivity law for polyhedral sets used in the proof of Proposition 2.5.1.

Corollary 4.3.1. *For every triplet of polyhedral sets P, Q, R we have*

$$P \times (Q + R) \cong (P \times Q) + (P \times R)$$

and

$$P \times 0 \cong 0$$

Thus it makes sense to talk about the Burnside rig of bounded polyhedral sets. A natural question we would like to answer is the following.

Question 2

What can we say about $\mathcal{B}(\mathcal{P})$? Is there an explicit description of it?

To begin with, it would be useful to see examples of equations that hold true in $\mathcal{B}(\mathcal{P})$. As we've observed in Example 4.2.1, by calling x the open interval $(0, 1)$, holds the following

$$x \cong 2x + 1 \tag{4}$$

So, even if a “ -1 ” object cannot exist by Remark 3.4.1, our x comes as close as possible. Developing this intuition, equation 4 tells us that, by evaluating

X in x , we have a natural map from the free polynomial rig over the relation $X \sim 2X + 1$ in $\mathcal{B}(\mathcal{P})$. Schanuel's theorem states that there is nothing else, answering Question 2.

Theorem 4.3.2 (Schanuel, 2006). *The canonical map*

$$\Phi : \frac{\mathbb{N}[X]}{X \sim 2X + 1} \longrightarrow \mathcal{B}(\mathcal{P})$$

is an isomorphism of rigs.

The following pages will be devoted to proving this theorem. Since the category of rigs is balanced, this will amount to proving injectivity and surjectivity of Φ .

Before proving the theorem, we state a simple but useful characterization of polyhedral isomorphisms involving open simplices, which are - as the name suggests - among the simplest polyhedra one could consider. We quickly recall their definition.

Definition 4.3.1. *We call **d -simplex** the convex hull of any $d + 1$ affinely independent points in some $\mathbb{R}^n$ with $n > d$. We call **open simplex** the topological interior of a simplex.*

It's not difficult to see that every d -simplex is affinely isomorphic to each other, and thus it is isomorphic to the *standard* d -simplex

$$\Delta_d = \left\{ \mathbf{x} \in \mathbb{R}^{d+1} \mid \sum x_i = 1, x_i \geq 0 \right\} = \text{conv}(\mathbf{e}_1, \dots, \mathbf{e}_{d+1})$$

Analogously, every open d -simplex is affinely isomorphic to the standard open d -simplex

$$S_d = \left\{ \mathbf{x} \in \mathbb{R}^{d+1} \mid \sum x_i = 1, 0 < x_i < 1 \right\}$$

Lemma 4.3.2. *Two polyhedral sets P and Q are isomorphic if and only if they admit a common partition into open simplices.*

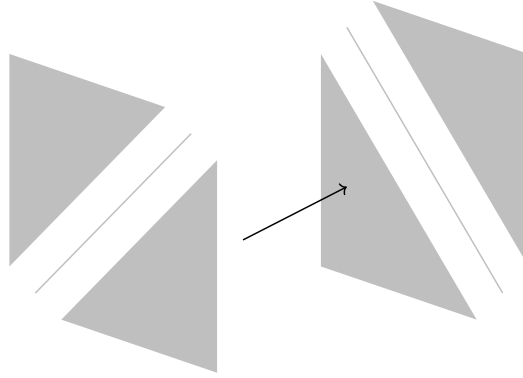
Proof. Clearly, two simplices are isomorphic if and only if they have the same dimension. Moreover, since piecewise-translations are polyhedral isomorphisms, splitting the polyhedral sets in their polyhedral partition is an isomorphism. Thus if they admit a common partition into simplices, they are isomorphic. Conversely, let f be a polyhedral isomorphism and let $P = \sum P_i$, $Q = \sum Q_i$ be partitions such that

$$f \upharpoonright P_i : P_i \longrightarrow Q_i$$

is an affinity for each i . Now, for every i , decompose P_i and Q_i in simplices

$$P_i = \sum_j R_{ij} \quad \text{and} \quad Q_i = \sum_k S_{ik}$$

Notice how these are, a priori, two different decompositions.

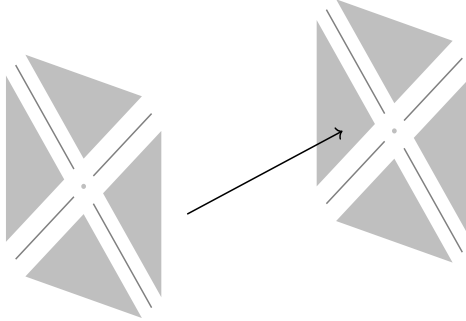


To get a common one, consider the partitions

$$P_i = \sum_k f^{-1}(S_{ik}) \quad \text{and} \quad Q_i = \sum_j f(R_{ij})$$

Since $\mathcal{P}$ is extensive, by Proposition 3.5.3 there exist a common refinement of the two partitions on P_i (or, equivalently, of the two partitions on Q_i):

$$P_i = \sum_{j,k} T_{ijk} \quad \text{and} \quad Q_i = \sum_{j,k} f(T_{ijk})$$



Now, up to decomposing T_{ijk} into simplices again, we are left with a common decomposition of P and Q in simplices. $\square$

A remarkable consequence of the previous lemma is the fact that, from a piecewise point of view, triangles are isomorphic to squares. This is actually one key lemma to prove the surjectivity of Φ .

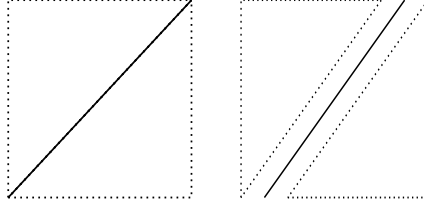
Lemma 4.3.3. *The open n -simplex S_n is isomorphic to the open n -cube x^n .*

Proof. The proof consists of an induction argument on n .

Base case Obviously the open interval $x = (0, 1)$ coincides exactly with the open 1-simplex S_1 .

Inductive case The strategy is to triangulate our n -cube to get an equation involving S_n .

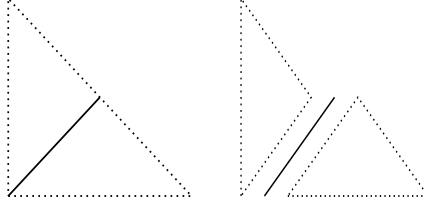
Let us start with an example in low dimension: take the open 2-cube (a square) and cut it along one of its diagonals.



This gives us an equation holding in $\mathcal{B}(\mathcal{P})$:

$$x^2 = 2S_2 + S_1$$

But we can observe that a similar decomposition holds for S_2 . Indeed, by cutting along a line from a vertex to its opposite edge,



we get the equation

$$S_2 = 2S_2 + S_1$$

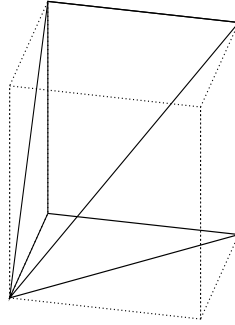
and thus, by Lemma 4.3.2, we can conclude that

$$x^2 = S_2$$

To generalize this proof in higher dimensions, one has to triangulate an n -cube into open n -simplices. Start with one vertex, $O = (0, \dots, 0)$ and consider the faces of our cube opposed to O , namely the hyperplanes given by the equation

$$x_i = 1 \quad \text{for } i = 1, \dots, n$$

Fix one of these faces and, for each of its vertices, draw a line to O . The result will be a n -dimensional pyramid with a $n - 1$ cube as its base.

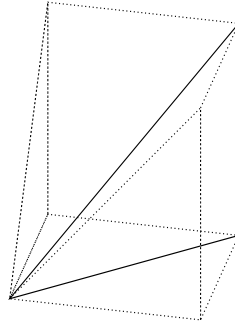


By repeating this process n times we partition the starting n -cube into n pyramids. But, by inductive hypothesis we know that

$$x^{n-1} = S_{n-1}$$

so our square base-pyramids are actually n -simplices, at least from the perspective of $\mathcal{P}$.

It's crucial to notice how, when working with *open* cubes and *open* simplices that this process only fills the n -dimensional volume of the cube but there are also some lower dimensional objects that this decomposition misses¹⁴. For example, take two different faces opposed to O : how many vertices do these two faces have in common? Since they have to satisfy two different equations $x_i = 1$ and $x_j = 1$ we have precisely 2^{n-2} points, i.e. an $n-2$ -cube. By connecting this cube (isomorphic to S_{n-2}) to O we get a $n-1$ -simplex unaccounted for in the previous decomposition.



Since this process holds true for every pair of faces, we have that the number of $n-1$ simplices appearing in our decomposition must be $\binom{n}{2}$. Our equation is slowly taking shape: for now we have

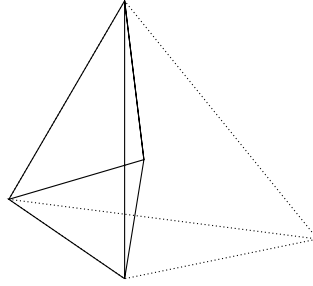
$$x^n = nS_n + \binom{n}{2}S_{n-1} + \dots$$

¹⁴If we didn't have these lower dimensional objects, we would've had the equation $x^n = nx^n$ which is obviously false.

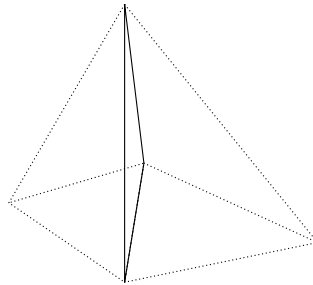
By iterating the same combinatorial reasoning as before one can show that the equation is precisely

$$x^n = \sum_{k=1}^n \binom{n}{k} S_{n-k+1}$$

To get a similar equation for S_n one has to triangulate it into open n -simplices, which can be done in the following way. Start with the usual vertex O and consider the centroid $G = (\frac{1}{n}, \dots, \frac{1}{n})$ on the face $F : \sum x_i = 1$ opposed to O . Now, fix a vertex on the face F and take the convex hull of all the other $n - 1$ vertices with G . This will be a $n - 1$ -simplex laying on the face F , so, by taking its convex hull with the point O we obtain an n -simplex.



By repeating this process for each vertex on F we get - just as before - n copies of S_n filling the volume of S_n . Once again, to count the $n - 1$ faces not captured by this process, start by asking what points do two faces have in common. Clearly, by excluding two vertices, one is left with $n - 2$ vertices which then form, by taking the convex hull with G and O , an $n - 1$ -simplex.



So, by repeating the same reasoning as before, we are finally left with the equation

$$S_n = \sum_{k=1}^n \binom{n}{k} S_{n-k+1}$$

And thus

$$x^n = S_n$$

□

For brevity's sake, we will call R the rig

$$R := \frac{\mathbb{N}[X]}{X \sim 2X + 1}$$

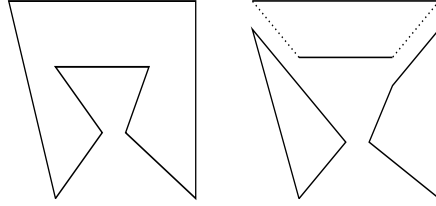
Exercise 4.3.1. Show that in R the equation

$$X^n = \sum_{k=1}^n \binom{n}{k} X^{n-k+1}$$

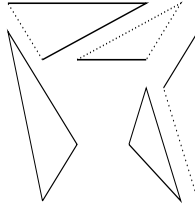
holds.

Corollary 4.3.2. *The map $\Phi : R \rightarrow \mathcal{B}(\mathcal{P})$ is surjective.*

Proof. Given a polyhedral set P , we can consider a decomposition into convex components. Notice how this decomposition is not generally unique but that's not needed right now.



Now, divide each convex component into open simplices.



And what it's left is a simple act of counting. Namely, consider the polynomial $P(X) \in R$ given by

$$P(X) = \sum_{i=0}^n a_i X^i$$

$a_i =$ number of open i -simplices in the decomposition of p .

For example, the polynomial associated to the polyhedral set pictured above is $P(X) = 5X^2 + 11X + 7$.

By Lemma 4.3.3, the open n -simplex S_n is isomorphic to the n -cube x^n which means that

$$\Phi(P(X)) = P$$

and thus Φ is surjective.

□

4.4 Euler characteristic and dimension of a rig

Injectivity is trickier. The key to prove it is to use algebraic variants of Euler characteristic and dimension - which we will denote by $\overline{\chi}$ and $\overline{\dim}$ - and check that those are injective. Defining $\overline{\chi}$ and $\overline{\dim}$ for R isn't particularly difficult, but - in order to generalize this proof to other future calculations - we will define them via universal properties, so that every rig can have its own Euler characteristic and dimension.

Definition 4.4.1. *Given a rig S we will denote with $E(S)$ the rig given by*

$$E(S) = S / \sim$$

$$\text{where } r \sim s \iff \exists t \in S \text{ such that } r + t = s + t$$

So basically we are freely forcing on S the cancellativity property. This will be the codomain of the Euler characteristic invariant. The definition for the analogous of the dimension invariant is a bit trickier:

Definition 4.4.2. *Given a rig S we will denote with $D(S)$ the rig given by*

$$D(S) = S / \approx$$

$$\text{where } r \approx s \iff r \leq s \text{ and } s \leq r$$

$$\text{where } r \leq s \iff \exists n \in \mathbb{N} \exists t \in S \text{ such that } r + t = ns$$

The intuition behind this condition is that if $r \leq s$ then “enough finitely many copies of s will absorb r ”. If the dimension of s is strictly less than the dimension of r this can never happen: it doesn't matter how (finitely) many lines you have, those will never cover a whole plane.

Proposition 4.4.1. $E(R) \cong \mathbb{Z}$

Proof. Very trivially, in R the equation

$$2X + 1 = X$$

holds, which means that in $E(R)$ the equation

$$X + 1 = 0$$

holds by cancellativity. Thus

$$E(R) = \frac{\mathbb{N}[X]}{X + 1 \sim 0} = \mathbb{Z}$$

□

Proposition 4.4.2. $D(R) = \{d^{-\infty}, d^0, d^1, \dots, d^n, \dots\}$ where $[0]_{\approx} = d^{-\infty}$, $[1]_{\approx} = d^0$ and the operations are given by

$$\begin{aligned} d^n + d^m &= d^{\max(n,m)} \\ d^n \cdot d^m &= d^{n+m} \end{aligned}$$

Proof. Firstly, let us show that the relation $\approx$ is well behaved relatively to the rig structure, i.e. that if $P \approx P'$ and $Q \approx Q'$ then

1. $P + Q \approx P' + Q'$
 2. $P \cdot Q \approx P' \cdot Q'$
1. Since $P \approx P'$ and $Q \approx Q'$, there exist $a, b, n, m \in \mathbb{N}$ and $U, V, S, T \in R$ such that

$$\begin{aligned} aP &= P' + U & bP' &= P + V \\ nQ &= Q' + S & mQ' &= Q + T \end{aligned}$$

without loss of generality we can suppose that $a, b, n, m \neq 0$. By using the equations on the left we get

$$\begin{aligned} an(P + Q) &= naP + anQ \\ &= nP' + nU + aQ' + aS \\ &= P' + Q' + (n-1)P' + (a-1)Q' + nU + aS \\ &\implies P' + Q' \leq P + Q \end{aligned}$$

Analogously by using the equations on the right one can show that also $P + Q \leq P' + Q'$, so

$$P + Q \approx P' + Q'$$

2. Similarly, by the equations on the left we can infer that

$$\begin{aligned} an(P \cdot Q) &= (P' + U) \cdot (Q' + S) \\ &= P'Q' + P'S + Q'U + US \\ &\implies P' \cdot Q' \leq P \cdot Q \end{aligned}$$

Analogously, we can show that also $PQ \leq P'Q'$, so

$$P \cdot Q \approx P' \cdot Q'$$

Let us go back to our description of $D(R)$; the thesis suggests we define

$$[0]_{\approx} = d^{-\infty} \quad [1]_{\approx} = d^0 \quad [X^n]_{\approx} = d^n$$

If we prove that for every polynomial $P \in R$ of degree n it holds that $P \approx X^n$, then the thesis follows from

$$\deg(X^n + X^m) = \max(n, m) \quad \deg(X^n \cdot X^m) = n + m$$

Once again, we prove this by induction on n .

Base Case: $X \approx aX + b$. Clearly, for every $a \neq 0$ it holds that $X \leq aX + b$ since

$$1 \cdot (aX + b) = X + (a - 1)X + b$$

Let us show the converse: recall that in R we have $X = 2X + 1$. By iterating b times, one get

$$\begin{aligned} X &= X + X + 1 \\ &= X + 2X + 1 + 1 = 3X + 2 \\ &= \dots = (b + 1)X + b \end{aligned}$$

If $b + 1 \geq a$ Then

$$\begin{aligned} X &= (b + 1)X + b \\ &= (a + b + 1 - a)X + b \\ &= aX + b + (b + 1 - a)X \end{aligned}$$

Since $b + 1 \geq a$, the term on the left is indeed an element of R and thus $aX + b \leq X$.

If $b + 1 < a$ Then

$$\begin{aligned} (a - b)X &= (a - b - 1)X + X \\ &= (a - b - 1)X + (b + 1)X + b \\ &= aX + b \end{aligned}$$

Where the first identity holds because both $a - b$ and $a - b - 1$ are natural numbers, and thus $aX + b \leq X$.

Inductive Case Let $P = \sum_{i=0}^n a_i X^i$ with $a_n \neq 0$. Using the base step on the tail of P in conjunction with item 1 we have

$$P = a_n X^n + \dots + a_1 X + a_0 \approx a_n X^n + \dots + X$$

Now, by the induction hypothesis and item 2, we can conclude that

$$P \approx a_n X^n + \dots + X = X \cdot (a_n X^{n-1} + \dots + 1) \approx X \cdot X^{n-1} = X^n$$

The only thing that is left to show is that these classes d^n are pairwise distinct, i.e. that $[X^n]_{\approx} \neq [X^m]_{\approx}$ for $n \neq m$. But this is easy: let us suppose that $n > m$, then if, by contradiction, there were some $T \in R$ such that

$$X^n + T = kX^m$$

we would have an equation where the term on the left is a polynomial of degree at least n while the term on right is a polynomial of degree m . Since our equation $X = 2X + 1$ leaves the degree untouched, this is clearly impossible even in R , thus ending our proof. $\square$

We are finally ready to define our invariants $\overline{\dim}$ and $\overline{\chi}$.

Definition 4.4.3. We call algebraic Euler characteristic (denoted by $\overline{\chi}$) the projection map from R onto $E(R)$

$$\begin{aligned}\overline{\chi} : R &\longrightarrow \mathbb{Z} \\ X &\longmapsto -1\end{aligned}$$

and we call algebraic dimension (denoted by $\overline{\dim}$) the projection map from R onto $D(R)$

$$\begin{aligned}\overline{\dim} : R &\longrightarrow D(R) \\ P &\longmapsto \deg(P)\end{aligned}$$

Now, to close the picture, we should prove that their geometric counterparts, χ and $\dim$, are polyhedral invariants so that they are well defined on $\mathcal{B}(\mathcal{P})$.

Proposition 4.4.3. If P and Q are two isomorphic polyhedral sets, then

$$\chi(P) = \chi(Q) \quad \text{and} \quad \dim(P) = \dim(Q)$$

Moreover, χ and $\dim$ are rig morphisms.

Proof. Since P and Q are isomorphic, by Lemma 4.3.2 they admit a common partition into simplices

$$P = \sum_i^n a_i S_i = Q$$

Thus

$$\dim(P) = n = \dim(Q)$$

and

$$\chi(P) = \sum_i^n a_i \chi(S_i) = \chi(Q)$$

As for the second part of the thesis, we already know that χ is a rig morphism from Proposition 2.5.1 so we only need to show it for $\dim$. Let P be as above and let $P' = \sum_j^m b_j S_j$ be another polyhedral set. Then

$$\begin{aligned}\dim(P \times P') &= \dim\left(\left(\sum_i^n a_i S_i\right) \times \left(\sum_j^m b_j S_j\right)\right) \\ &= \dim\left(a_n b_m (S_n \times S_m) + \dots\right)\end{aligned}$$

But by Lemma 4.3.3 we get

$$\begin{aligned}\dim(P \times P') &= \dim\left(a_n b_m (S_n \times S_m) + \dots\right) \\ &= \dim(a_n b_m (x^n \times x^m) + \dots) \\ &= \dim(a_n b_m (x^{n+m}) + \dots) = n + m \\ &= \dim(P) + \dim(P')\end{aligned}$$

Since the sum is the rig product in $D(R)$, $\dim$ sends product into product and is thus a rig morphism. $\square$

4.5 A full classification of bounded polyhedra

As of now the picture is

$$\begin{array}{ccc} R & \xrightarrow{(\overline{\chi}, \overline{\dim})} & E(R) \times D(R) \\ & \searrow \Phi & \nearrow (\chi, \dim) \\ & \mathcal{B}(\mathcal{P}) & \end{array}$$

It's easy to see that, given our definitions, this diagram commutes. Our efforts in the next pages will be devoted to showing that the map above $((\overline{\chi}, \overline{\dim}))$ is injective, which will then force Φ to be injective too, and thus an iso.

Theorem 4.5.1. *The map*

$$(\overline{\chi}, \overline{\dim}) : R \longrightarrow E(R) \times D(R)$$

is injective.

Proof. Let $P, Q \in R$ be such that

$$\overline{\chi}(P) = \overline{\chi}(Q) \quad \text{and} \quad \overline{\dim}(P) = \overline{\dim}(Q)$$

Notably, since $\dim(P) = \dim(Q)$, P and Q have the same degree so we can write them as

$$P = \sum_{i=0}^n a_i X^i \quad Q = \sum_{j=0}^n b_j X^j$$

The proof consists - once again - of an inductive reasoning on the degree n .

Base Case Let $n = 1$ so that $P = a_1 X + a_0$ and $Q = b_1 X + b_0$ with $a_1, b_1 \neq 0$. The condition $\chi(P) = \chi(Q)$ becomes

$$-a_1 + a_0 = -b_1 + b_0$$

Clearly, if $a_1 = b_1$ the condition above ensures that $a_0 = b_0$ too, from which the thesis follows. So let us suppose that $a_1 \neq b_1$, and, without loss of generality, that $b_1 > a_1$.

Starting with P , we can make a chain of equalities using the equation

$$X = 2X + 1:$$

$$\begin{aligned}
P &= a_1X + a_0 = (a_1 - 1)X + X + a_0 \\
&= (a_1 + 1)X + a_0 + 1 \\
&= \dots \text{ iterating } b_1 - a_1 \text{ times } \dots \\
&= (a_1 + b_1 - a_1)X + a_0 + b_1 - a_1 \\
&= b_1X + b_1 + (a_0 - a_1) \\
&= b_1X + b_1 - b_1 + b_0 \\
&= b_1X + b_0 = Q
\end{aligned}$$

Hence the thesis.

Inductive Case Let $P = \sum_{i=0}^n a_i X^i$ and $Q = \sum_{j=0}^n b_j X^j$ with the condition

$$\sum_{i=0}^n (-1)^i a_i = \sum_{j=0}^n (-1)^j b_j$$

Without loss of generality we can suppose that $b_0 \geq a_0$, say $b_0 = a_0 + k$ for some $k \in \mathbb{N}$ (notice how k can be zero).

With the same trick as before we can equalize the constant terms:

$$P = a_n X^n + \dots + a_1 X + a_0 = a_n X^n + \dots + (a_1 + k)X + b_0$$

Define the polynomials P' and Q' of degree $n - 1$ as follows:

$$P' = \sum_{i=0}^{n-1} a_{i+1} X^i + k \quad \text{and} \quad Q' = \sum_{j=0}^{n-1} b_{j+1} X^j$$

so that we have

$$P = X \cdot P' + b_0 \quad \text{and} \quad Q = X \cdot Q' + b_0$$

Notice that P' and Q' not only have the same algebraic dimension ($n - 1$) but also the same algebraic Euler characteristic. Indeed,

$$\bar{\chi}(P') = -\bar{\chi}(P) + b_0 = -\bar{\chi}(Q) + b_0 = \bar{\chi}(Q')$$

Hence, by inductive hypothesis, $P' = Q'$ which also forces P and Q to be equal, ending our proof.

□

Corollary 4.5.1 (Main Theorem). *The map $\varphi : R \rightarrow \mathcal{B}(\mathcal{P})$ is an isomorphism, and two polyhedral sets P and Q are isomorphic if and only if*

$$\chi(P) = \chi(Q) \quad \text{and} \quad \dim(P) = \dim(Q)$$

Remark 4.5.1. After proving the injectivity of the map $(\overline{\chi}, \overline{\dim})$, one could try their hand at proving surjectivity too. Sadly, true surjectivity is out of bounds. In fact, if $(\overline{\chi}, \overline{\dim})$ were surjective, one would have a polynomial (and a polyhedral set too since now we've proven that they are one and the same!) whose degree is 0 and whose Euler characteristic is -1 . Such a polyhedral set would then be a “negative point” and we've already seen in Remark 3.4.1 how this cannot happen.

Now, while true surjectivity is unreachable, again our $\mathcal{B}(\mathcal{P})$ comes as close as possible: for every pair $(z, d^n) \in E(R) \times D(R)$ outside of the case $z < 0 \wedge n = 0$ there exists a polynomial P such that $\overline{\chi}(P) = z$ and $\overline{\dim}(P) = d^n$. Indeed, one can construct a polynomial $P \in R$ depending on the values of z and n in the following way:

$$P = \begin{cases} zX^n & \text{if } z \in \mathbb{N} \text{ and } n \text{ is even} \\ X^n + (1 - z)X^{n-1} & \text{if } z \notin \mathbb{N} \text{ and } n \text{ is even} \\ -zX^n & \text{if } -z \in \mathbb{N} \text{ and } n \text{ is odd} \\ X^n + (z - 1)X^{n-1} & \text{if } -z \notin \mathbb{N} \text{ and } n \text{ is odd} \end{cases}$$

Making these the only four possible polyhedral sets, up to isomorphism.

5 Ringraziamenti

Ci sono categoristi¹⁵ che interpretano il Lemma di Yoneda come l'affermazione che ogni individuo è completamente determinato come persona dalle sue interazioni con gli altri. Nonostante pensi che l'atto di generalizzare formule matematiche a massime filosofiche romantiche sia pericoloso e un po' tacky¹⁶, devo dire che mi ritrovo molto in questa massima. In quest'era di individualismo pazzo e sfrenato, di "ognuno per sé", di produttività e # on the grind, io sono contento di affermare che sono arrivato dove sono arrivato solo per il continuo e incrollabile supporto delle persone intorno a me. Questa paginetta di ringraziamenti è per voi, nella speranza di restituirvi almeno una briciola di quanto mi avete dato.

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¹⁵brutte bestie, lo so.

¹⁶ci ricordiamo tutti i tatuaggi con la formula di dirac, sì?

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